Learning Whom to Trust from Feedback: How History Format Shapes In-Context Learning in Vision-Language Models

Anonymous submission

Abstract

Learning whom to trust when an advisor's accuracy changes is a classic test of context-dependent learning in people. We adapt a human advice-taking task to ask whether vision-language models (VLMs), without parameter updates, learn advisor accuracy in context from trial-by-trial feedback, and how this depends on the format of the history. The same history is given raw (both reports and the outcome), as a record of who was correct on each trial, or as counts of correct trials. On 48 held-out episodes, four of five open-weight VLMs answering directly chose at or below chance from raw histories and tended to repeat the previous outcome. A record of who was correct raised accuracy by 8–16 percentage points, to the level of the counts, whereas appending the same record to the raw history helped little; masking parts of the history showed that the visible outcome label drives the repetition. The effect depends on the system and the layout: two proprietary reasoning models, tested on a subset, had similar observed raw-history accuracy to a Bayesian volatility learner (.63), and a compact one-line layout narrowed the gap in two open models. In a changing bandit from Psych-101, a dataset of human choices, participants' estimated forgetting increased with change frequency, while model estimates changed little when replaying participants' histories. The task and analysis code will be released upon acceptance.

Introduction

People learn whom to trust from feedback. When two advisors disagree and the outcome is revealed afterwards, people keep track of how often each advisor was right, and they update this estimate faster when the advisor's accuracy is volatile (Behrens et al. 2008); advisor reputations form quickly from a track record (Yaniv and Kleinberger 2000; Bonaccio and Dalal 2006). Because this advice-taking task comes from human neuroscience, a model can be tested with the same design and analysed with the same computational models. We use a simplified variant without human data; our direct comparison with people uses a second task with published human choices.

Agents built on large language models (LLMs) and vision-language models (VLMs) face the same problem when they combine tool outputs, retrieved documents or messages from other agents whose reliability can change. Prior work asked whether LLMs weight sources by reliability when it is fixed (Ashkinaze et al. 2026), flips once (Zhou et al. 2026) or breaks after scripted failures (Chen 2026), and proposed memories that track reliability outside the model (Feng, Yang, and Poria 2026). Whether a model without fine-tuning can learn a changing, non-stationary accuracy by in-context learning from trial-by-trial feedback, and which part of the computation fails if it cannot, is open. This is context-dependent learning without parameter updates: each query receives the accumulated history, and the model's choices do not change later feedback.

The format of past evidence is known to matter. LLMs choose differently when outcome probabilities are described than when they are experienced (Ge, Zhang, and Vorobeychik 2026); summarized histories can improve exploration in bandits (Krishnamurthy et al. 2024), and evidence integration can show strong recency (Xue, Wilson, and Xiong 2026). These tasks do not require deriving advisor correctness by comparing conflicting reports with observed outcomes.

Learning whom to trust involves three computations: (i) judging which advisor was correct on each trial, (ii) integrating these judgments across trials, and (iii) using the estimate. Five VLMs receive the same history in formats that state these results to different degrees (Fig. 1b), plus a condition describing advisor accuracy. Formats also change length and salience; we test which representations support good choices without assuming separable internal stages. Our contributions are:

- An advice-taking evaluation adapted from human neuroscience, with text and image reports; task generation, prompts and analysis code will be released upon acceptance.
- Across five open VLMs answering directly, correctness-only histories remove most of the summary–raw gap, whereas a visible outcome label elicits repetition of the previous outcome, as predicted before a held-out masking test.
- A boundary condition: two proprietary reasoning models have similar observed raw-history accuracy to a Bayesian learner, so the format effect depends on the system.
- A human–model comparison in Psych-101.

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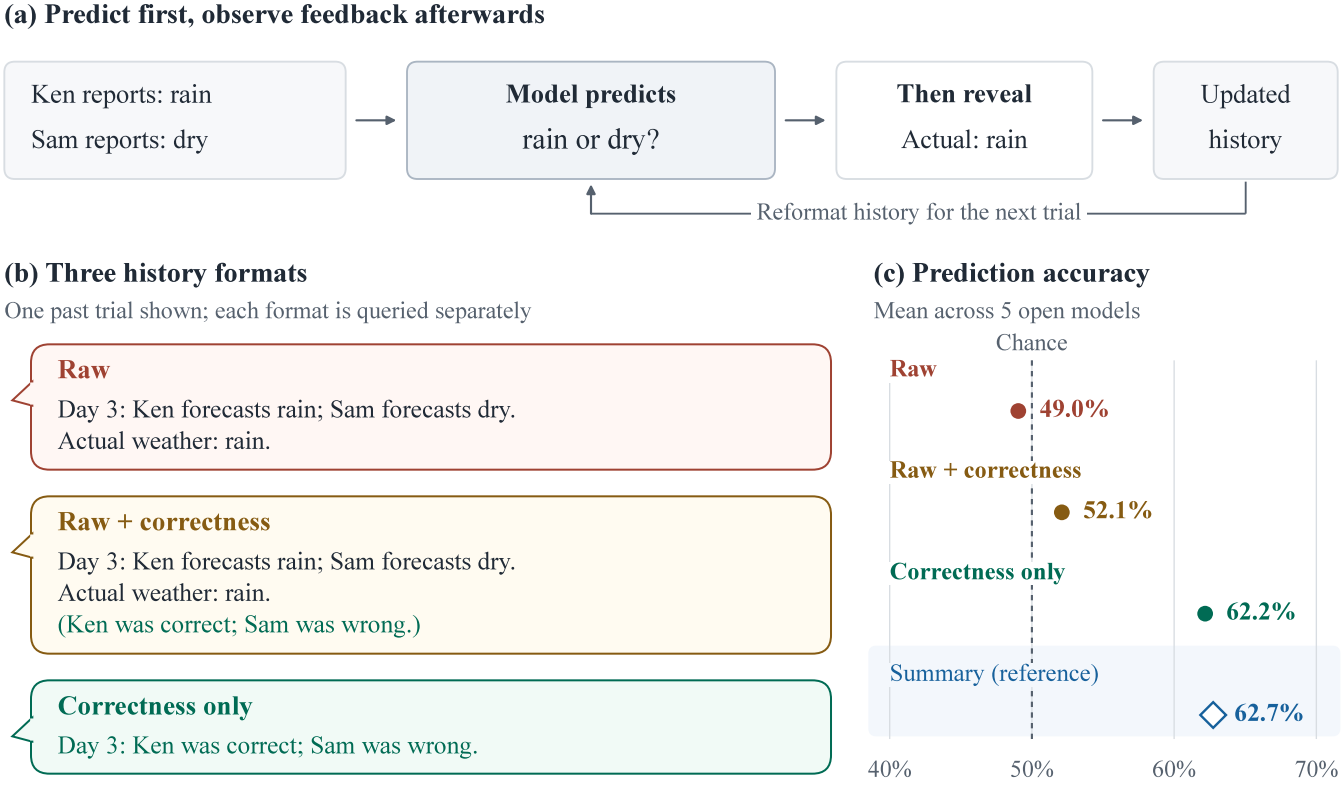


Figure 1: Replacing raw feedback with correctness-only histories improves prediction accuracy. (a) Predictions use past history and current reports; the current outcome is revealed afterwards and enters the next trial's history. (b) Prompt excerpts show the same past trial in three history formats, supplied in separate queries. (c) Equal-weight mean accuracies over five open models on 48 held-out test episode pairs with text reports. A pair comprises an episode and its counterbalanced copy, with advisor names and answer labels swapped. The open diamond marks all-trial summaries; the dashed line marks 50% chance. Table 1 gives model-level accuracies and paired 95% bootstrap confidence intervals (CIs) for summary−raw differences; Table 7 gives the individual accuracy CIs.

Related Work

Cognitive evaluation of LLMs. Treating a language model as a participant in a psychological experiment is now a common way to characterize its behavior (Binz and Schulz 2023). CogBench turns such experiments into behavioral metrics (Coda-Forno et al. 2024), and Centaur is a model fine-tuned to predict trial-by-trial human choices in Psych-101 (Binz et al. 2025). Fits of reinforcement-learning models to in-context behavior in bandits have found asymmetric or biased updating and rigid or greedy choice (Schubert et al. 2024; Hayes, Yax, and Palminteri 2024; Wang, Ward, and Zhang 2026b; Schmied et al. 2025), and reversal-learning schedules have been compared across LLMs (Wang, Ward, and Zhang 2026a; Li et al. 2025). The bandit and reversal-learning evaluations focus on learning option values rather than extracting advisor correctness from report–outcome pairs.

Description, experience and history summaries. The description–experience gap (Hertwig et al. 2004; Wulff, Mergenthaler-Canseco, and Hertwig 2018) has an LLM counterpart in the description–history gap of Ge, Zhang, and Vorobeychik (2026). Krishnamurthy et al. (2024) contrasted raw and summarized histories in bandits, and Xue, Wilson, and Xiong (2026) traced integration kernels (weights on evidence at different lags) in a counting task. LLMs also update roughly as Bayesian observers when the observed sequence is itself what they predict (Gupta et al. 2025; Zhang, Yang, and Wang 2025). Our summarized history is computed from the same history the model would otherwise see; it discards trial order and is not a description. Source-history interventions also distinguish recognizing reliability from allowing it to influence a decision (Zhang, Jiang, and Lu 2026). We extend this question to repeated reversals and test which components of trial-by-trial feedback trigger outcome repetition.

Volatility and multimodal reports. Our task follows the social-learning paradigm of Behrens et al. (2008). Normative accounts of learning under volatility include hierarchical Bayesian and change-point models (Behrens et al. 2007; Nassar et al. 2010, 2012; Mathys et al. 2011; Piray and Daw 2021, 2024; Soltani and Izquierdo 2019), and social and individual learning can be arbitrated separately (Diaconescu et al. 2020). For image reports, performance depends on modality

even when content is isomorphic (Fu et al. 2024), multi-image inputs show position biases (Tian et al. 2025), and VLMs have difficulty binding features to objects (Campbell et al. 2024).

Task and Methods

Advice-taking task. On each trial, two advisors give conflicting binary reports (for example, rain and dry), the model predicts the outcome, which amounts to choosing whom to follow, and the outcome is then revealed. An episode has 80 trials. Advisor A is correct with probability 0.8 in a stable phase of 40 trials; in a volatile phase of 40 trials its accuracy alternates between 0.8 and 0.2 every 10 trials (each switch is a reversal), following Behrens et al. (2008) (comparison in Appendix B). Half of the episodes start with each phase. Outcome sequences are pre-generated, so the model's choices do not change the history. Within each 10-trial block A's realized accuracy equals the scheduled value, and each outcome label appears five times. The instructions say that an advisor's accuracy may change but give no rate.

Each episode is shown with a pet cover story, in which reporters say whether a hidden animal is a cat or a dog, and with one of six non-pet cover stories (colors, weather, stocks, sports, medical tests, news; Appendix A). Current reports are given as text or as an image: a drawn icon for the non-pet cover stories and a photograph (Parkhi et al. 2012) for the pet cover story; with text reports, both cover stories are text only. Every episode has a counterbalanced copy in which advisor names and answer labels are swapped while outcomes are identical; averaging over the two cancels name and label biases. The 16 development episodes were used to choose settings; 48 separate episodes form the held-out test set. A robustness set has 24 episodes with independent outcomes and irregular reversals.

History formats. Before trial t, the model sees trials $1, \ldots, t-1$ in one format (examples in Fig. 1b; full prompts in Appendix A). The *raw history* lists both reports and the outcome ("Day 3: Ken forecasts rain; Sam forecasts dry. Actual weather: rain."). *Raw + correctness* appends "(Ken was correct; Sam was wrong.)" to each trial. *Correctness only* keeps only that statement, stating the result of step (i). The *summary* gives each advisor's number of correct trials so far, stating the result of steps (i) and (ii); it discards trial order. Further summaries count over the last K trials or since a change point inferred by Bayesian online change-point detection (BOCPD; Adams and MacKay 2007; Appendix E), and a chart draws who was correct as an image.

Control conditions. On development episodes, control items give the answer more directly: *described accuracy* ("Ken was correct on 80% of past days"), described counts, short and long easy histories in raw and correctness-only form, and a single raw trial on which the model names who was correct (step (i) alone).

Models and readout. We test five open-weight VLMs without fine-tuning, four of them built on Qwen3 language models: Qwen3-VL 8B and 32B (Bai et al. 2025), InternVL3.5 8B and 38B, which use a Qwen3 language model (Wang et al. 2025), and Gemma-3-27B (Gemma Team 2025). The open models are prompted to answer directly with one word. Two proprietary reasoning models are tested through application programming interfaces (APIs) on a subset; their choice is read from the generated answer (Appendix J). The readout is the choice probability y_t for advisor A's report, from the logits (unnormalized output scores) of the two answer words at the first generated token, normalized over these two words; it is not a belief. For four models nearly all probability mass fell on the two answer words; Gemma-3-27B often began a sentence instead when given a summary (Appendix J).

Measures and analysis. Accuracy is the proportion of trials ($t \geq 2$) on which the higher-logit answer is correct, after averaging label-aligned logits over the counterbalanced copies; exact ties receive half credit and chance is 0.5. To describe behavior we fit a logistic regression of y_t on which advisor was correct k trials back, whose weights over k form an integration kernel (cf. Xue, Wilson, and Xiong 2026), plus a regressor for repeating the previous outcome, and compute agreement (Cohen's κ) with simple strategies. As a computational reference we run a hierarchical Bayesian volatility learner inspired by Behrens et al. (2007, 2008) (implementation in Appendix D). It receives who was correct on each trial and assumes conditionally independent Bernoulli observations, without modelling our block quotas. The unit of analysis is the episode pair (an episode and its counterbalanced copy), and uncertainty is reported as 95% confidence intervals (CIs) from a bootstrap over episode pairs (10,000 resamples). These CIs are pointwise: they are not adjusted for simultaneous coverage across estimates. The primary test, fixed before held-out evaluation, is the summary−raw accuracy difference per model (Wilcoxon signed-rank test, Holm-corrected across models; required in both language-model families). Secondary tests and all other details are in Appendix C.

Human comparison. We replay the changing-bandit task of Xiong et al. (2023) from Psych-101: 30 participants, three change probabilities (hazard rates, h) and 100 trials per game. The hazard rate is the probability that a machine's payoff changes on each trial. The five open models see each participant's preceding choices and payoffs, with h hidden or stated. We fit an adapted myopic model-free learner to human choices and sampled model choices. Its forgetting rate ϕ moves the unchosen machine's estimated value toward the midpoint of the payoff range; we compare slopes of ϕ on h, paired by participant (Appendix H).

Results

Open models use described accuracy more reliably than raw feedback. With text reports, described accuracy or counts led the models to choose the advisor described as better on 92–100% of items (a different measure from main-task accuracy), and every model named the correct advisor on a single shown trial (Table 6; development episodes). The same information was used much less when it had to be learned. In an easy raw history in which one advisor had been right on every recent trial, models followed that advisor

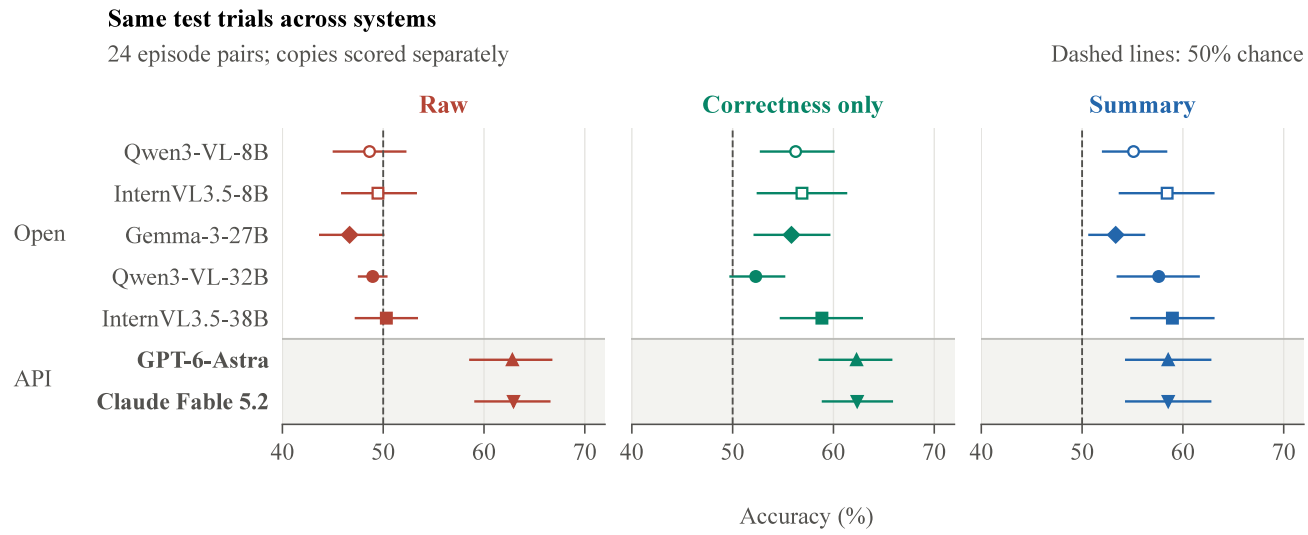


Figure 2: History-format effects differ across systems on the same test trials. Seven systems are compared on 24 held-out episode pairs, every fourth trial, with each counterbalanced copy scored separately. Open-model choices use candidate logits; API choices use generated answers (valid answers only). Each panel shows one history format. Bars show 95% bootstrap CIs from resampling episode pairs; dashed lines mark 50% chance. The comparison does not isolate reasoning from other system differences. Full-set results appear in Table 1.

on .60–.79 of items after 10 trials and on .53–.70 after 80. Stating who was correct on each trial raised this to 1.00 for 10 trials, but only to .67–.90 for 80 trials. The difficulty therefore emerges when using multiple feedback trials, and long histories remain hard even when correctness is stated explicitly.

Correctness-only histories recover most of the gap; appending correctness helps less. On held-out episodes, accuracy was .468–.486 with raw histories in four models and .549 [.538, .560] in InternVL3.5-38B, the only model above chance, and .617–.635 with summaries (Table 1; individual accuracy CIs in Table 7). The summary−raw gap was $+.084$ to $+.164$ and significant in every model (Holm-corrected $p < .001$; Table 1), so the primary test held in both language-model families. Scoring each counterbalanced copy separately preserved positive gaps in all five models ($+8.8$ to $+13.9$ points; Appendix J). A correctness-only history attained .602–.636 accuracy, close to the summarized history, although it still leaves integration to the model. Integration was nonetheless imperfect: this history remained below a count over the last five trials (.692) and lagged after reversals (below). Appending the same statement to the raw history improved accuracy by only 1.1–5.4 percentage points, even though the model then had all the information of the correctness-only history. The raw reports and outcomes therefore do more than add work: they interfere with using the statement next to them, which the next analyses explain. The size of the gap depends on presentation: with a compact one-line record per trial, raw-history accuracy rose to .55–.56 while correctness-only and summarized histories fell, shrinking the gap to 1–3 points in the two 8B models; raw accuracy stayed below the count rule in every layout (development episodes; Appendix I).

Under raw histories, open models tend to repeat the previous outcome. The kernel weights on past correctness were near zero, while the weight on repeating the previous outcome was positive in every model (Fig. 8; correctness kernel in Fig. 3). Repeating the previous outcome agreed best with the models' choices ($\kappa = .22$–$.32$ in four models and .13 in InternVL3.5-38B, whose choices also agreed weakly with recent counts; Fig. 9). The 15-lag kernel describes local sensitivity to recent trials and cannot by itself separate a recency-weighted learner from an all-trial count on this schedule; identifiability checks are described in Appendix F. Because each 10-trial block contains each outcome five times, repeating the previous outcome scores below chance (.445 on held-out episodes), which explains accuracy below .5; on the robustness set, where outcomes are independent, raw-history accuracy was near chance (.507–.520) in four models. When the history stated who was correct, the kernel instead peaked at the most recent trial and decayed over the next few, and the weight on the previous outcome vanished.

Masking identifies the outcome label as a trigger. We replaced parts of the raw history with correctness by a dash while retaining its layout. Development results motivated predictions for a five-model held-out replication (Appendix I). On held-out episodes, all predictions written before the run held in all five models: replacing the outcome label helped by 2.7–6.4 points and reduced the weight on the previous outcome from .42–.87 to .08–.23, replacing the reports alone changed accuracy by at most 0.4 points, but once the outcome label was replaced, also replacing the reports added 2.1–5.5 points, and with both replaced accuracy came within 4 points of the correctness-only history (Table 2; Ap-

Model	Raw	Raw+corr.	Corr. only	Summary	Summary−raw [95% CI]	p_{Holm}
Full test set: 48 episode pairs; copies combined before scoring						
Qwen3-VL-8B	48.6	51.7	61.9	62.1	+13.4 [12.2, 14.6]	<.001
InternVL3.5-8B	47.9	50.8	63.6	63.5	+15.6 [14.5, 16.7]	<.001
Gemma-3-27B	47.0	49.8	60.2	61.7	+14.7 [13.2, 16.3]	<.001
Qwen3-VL-32B	46.8	52.2	62.8	63.1	+16.4 [15.0, 17.7]	<.001
InternVL3.5-38B	54.9	56.0	62.5	63.3	+8.4 [7.4, 9.4]	<.001
API subset: 24 episode pairs, every fourth trial; copies scored separately						
GPT-6-Astra	62.8	–	62.3	58.5	−4.3 [−8.8, 0.3]	–
Claude Fable 5.2	62.9	–	62.4	58.5	−4.4 [−8.6, 0.1]	–
Computational references: full set / API subset						
Bayesian learner		66.8 / 62.9			–	–
All-trial count		63.2 / 58.5			–	–
Last-five count		69.2 / 67.9			–	–

Table 1: Correctness-only histories raise accuracy in the five open models. Accuracy is in percent; differences and their paired 95% bootstrap CIs are in percentage points. Full-set scores average the two cover stories after aligning and averaging the counterbalanced logits. API scores use generated answers on a smaller set of trials and are reported separately; Fig. 2 compares all systems on the same trials with per-copy scoring. Raw+corr.: raw history with correctness appended. Primary p-values are Holm-corrected across the five open models; secondary tests and controls appear in Appendix C; multimodal checks appear in Appendix G.

	Outcome shown		Outcome masked	
Model	Reports shown	Reports masked	Reports shown	Reports masked
Qwen3-VL-8B	51.7	52.0	55.7	60.4
InternVL3.5-8B	50.8	50.4	56.1	59.6
Gemma-3-27B	49.8	49.9	55.0	60.5
Qwen3-VL-32B	52.2	51.8	58.6	60.7
InternVL3.5-38B	56.0	55.7	58.6	63.3

Table 2: Masking past outcomes improves accuracy; masking reports as well yields further gains. Accuracy (%) on 48 held-out episode pairs with text reports and counterbalanced logits combined before scoring. Columns specify whether past outcomes and advisor reports are shown or replaced by a dash; correctness statements remain visible in all conditions. The first column is raw + correctness. Paired changes and their 95% bootstrap CIs appear in Fig. 11.

pendix I). The visible outcome label drives the repetition; the reports matter once it is removed.

Reliability judgments and choices are weakly aligned. Asked which advisor was currently more likely to be correct, the models named the scheduled better advisor on .60–.67 of raw-history trials, below always naming the advisor that was better over most of the episode (.75), with little agreement with any rule ($\kappa \leq .17$), and their choices followed their own judgment on only 53–69% of trials ($\kappa \leq .12$; Appendix I). Judgments and choices were weakly aligned.

A computational reference for evidence integration. The Bayesian reference learner, run on the same held-out episodes, made correct choices on .668 [.661, .674] of trials, above the all-trial count rule (.632) and above every open model given raw, correctness-only or all-trial summarized histories (Table 1; Appendix B). It receives who was correct on each trial, so it benchmarks integration separately from reading the history, and it is not human performance. A count over the last five trials, selected on development episodes with the same schedule, scored higher (.692), and only models given such a recent-trial summary, computed outside the model, reached the learner (.666–.691; Appendix E). With 80 trials, even this learner's learning rate rises only slightly in the volatile phase, too little to test volatility adaptation here (Appendix F).

A summary passes on its rule, including its lag. Given all-trial counts, models chose the advisor with more correct trials on 94–100% of non-tied trials, and four of five were equivalent to the rule alone within ± 2 points (Gemma-3-27B: $-.014$, $p_{\text{Holm}} = .053$; Table 5). A model that follows a summary therefore inherits the rule's response to change. The all-trial count lags behind every reversal (Appendix B, Fig. 6): after a reversal, the mean probability of following the currently better advisor stayed below .5 early in the block with either this summary or the correctness-only history, then rose above .5; with raw histories it stayed near .50 (Fig. 4). Summaries over the last five trials or since a change point inferred by BOCPD were followed as faithfully (agreement .94–1.00; equivalent to the rule in five of five models for the change point and three of five for the window), and the mean following probability exceeded .5 two to three trials after a reversal; overall accuracy was .666–.691, 4–6 points above the all-trial summary and 4–8 points above the correctness-only history (Appendix E). How fast an agent built this way adapts to a volatile source is therefore set by the rule outside the model.

Proprietary reasoning models learn from raw histories. On a held-out subset (24 episode pairs, every fourth trial; Appendix J), Claude Fable 5.2 made correct choices on

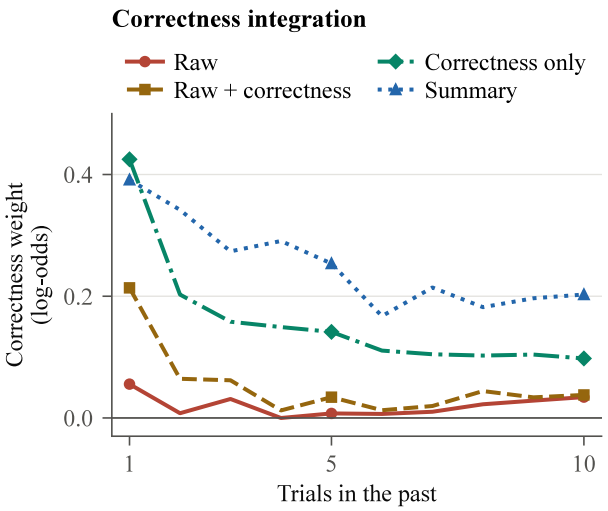


Figure 3: Correctness-only and summarized histories elicit decaying weights on past correctness. Curves show correctness integration kernels by history format, averaged over the five fixed open models on held-out test episodes with text reports. The first 10 of 15 fitted lags are shown; line styles, markers and colours identify formats. Per-model kernels and CIs appear in Fig. 8, and raw-history strategy agreement in Fig. 9.

.629 [.592, .665] of trials with raw histories, .624 with the correctness-only history and .585 [.544, .627] with the all-trial summary (summary−raw −.044 [−.086, .001]); GPT-6-Astra gave nearly identical values (.628, .623 and .585; −.043 [−.088, .003]). On these same trials, with each counterbalanced copy scored separately, the five open models reached .467–.503 with raw histories, .523–.589 with correctness-only histories and .533–.590 with summaries (Fig. 2); the Bayesian learner reached .629 and the count rules .585 (all trials) and .679 (last five). Correctness labels did not increase mean accuracy for these models, and the all-trial summary was numerically lower than the raw history, at the level of its rule. These models reason before answering (visible reasoning in 77% of Fable's raw answers) and also differ from the open models in scale, training and readout, so the comparison sets a boundary on the format effect rather than isolating an effect of reasoning. Their raw-history choices were more consistent with advisor tracking than with the tested shortcuts (exploratory; Appendix J): they agreed with the Bayesian learner (κ = .78–.87) and the last-five count (.77–.79), not with repeating the previous outcome ($\kappa \approx 0$) or with the outcome quota (.07–.10), a rule that would score .702 on these trials; where the quota and the learner disagreed they followed the learner on 91–96% of trials, and after a reversal they followed the newly better advisor on .06–.09 of trials in the first three trials of the reversed block, rising to .94–.99 in trials 8–10. Given the all-trial summary, both chose exactly as the count rule ($\kappa = 1.00$).

Image reports add context-dependent costs (exploratory). Models named a single report image almost perfectly, yet with described accuracy and image reports they followed the better advisor on only .59–.84 of items, in four models least when the advisor's name was given only after the image (Appendix G). Accuracy was mostly higher when the name was printed on the image or both reports were shown in one labelled image, although each panel was read correctly. On held-out episodes much of the cost came from choosing the first-listed image: with the correctness-only history, four models chose the first-listed report on 79–92% of trials with images but 44–54% with text, and their accuracy fell to .11–.24 when the correct report was listed second (Appendix I). With the correctness-only history, image reports lowered held-out accuracy by 7–10 points compared with text reports (.523–.552). Models could not read single cells of the correctness chart reliably (.61–.71), so chart histories are not interpreted.

Humans and open models in the changing bandit. In the changing bandit of Psych-101, a normative learner forgets the unchosen option faster when the hazard rate is higher, and participants did (median forgetting rate .079 at $h = 0.1$, .255 at $h = 0.4$; Appendix H). Unlike the advice task, this task varies volatility across games. The five open models replayed participants' histories; participants had also played earlier games, so the comparison is descriptive. A simulation of the whole pipeline gave a false-positive rate of 6% but only about half the power for a difference as large as the human effect (Appendix H), so a null result would not show similarity. Model estimates varied less with hazard rate. Their median forgetting rate was at most .051 at every hazard rate, and the per-participant slope on hazard rate was -0.01 to 0.04, against 0.47 for the participants (Fig. 5). All five paired differences were negative (-0.48 to -0.60), four at an uncorrected $p < .05$ (p = .011–.070), none after Holm correction across the five models. Stating the hazard rate, which matches what participants were told, did not produce a clear change (slopes -0.04 to 0.16). Forgetting rates near zero are estimated at the boundary of the parameter range, so we checked this in a separate regression: the models' choices did depend on the machines' last observed values (positive weight for 77–100% of participant histories), but whereas participants weighted a value difference less when the older value was stale (seen four or more trials earlier; stale/fresh weight ratio .58–.86), the models weighted stale differences as much or more (.95–2.07; Appendix H). This exploratory contrast does not establish a hazard-dependent adaptation deficit: the human discounting estimate did not increase with hazard rate, and replay differs from acting in the task.

Discussion

Across five open VLMs answering directly, history representation strongly affected how well models used advisor accuracy. The models used described accuracy and followed a summary as faithfully as its rule, yet four of five chose at or below chance from the raw record. A record of who was correct removed most of the gap, the same record appended to the raw history helped little, and masking showed that the

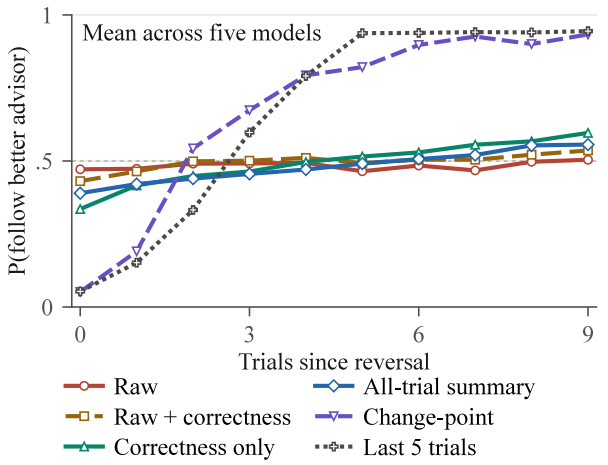


Figure 4: Recent-trial summaries recover higher following probabilities sooner after a reversal. Zero marks the first trial after a reversal. Curves are mean choice probabilities over the five fixed models on held-out episodes, with line styles and colours identifying history formats. The dotted horizontal line marks .5; pointwise uncertainty is not shown. Per-model curves appear in Fig. 7.

visible outcome label elicited repetition of the previous outcome. This suggests a refinement of the description–history gap (Ge, Zhang, and Vorobeychik 2026): the raw record promotes a competing outcome-repetition response even when trial-level correctness is explicitly stated. Our manipulations change what the history states, not internal stages, so they do not localize a failed internal computation. The effect is also not universal: two proprietary reasoning models had similar observed raw-history accuracy to the Bayesian learner, with no increase in mean accuracy from correctness labels. In the human study, who was correct is the observation from which a prediction error about the advisor is computed (Behrens et al. 2008); our interventions change how this observation is represented and make no claim about an internal error signal or its neural implementation.

Implications for NeuroAI. For building agents, an explicit record of which source was correct, and an integration rule outside the model, help models that struggle with raw histories: with a last-five-trial summary, mean following probability exceeded .5 three trials after a reversal; with all-trial counts it crossed .5 six trials after the reversal and reached only .56 at the end of the ten-trial window (Fig. 4). For models that already use raw histories, a lossy summary can instead remove information. For neuroscience, the format manipulation offers a way to test candidate models of advice taking against the behavioural signatures used for people; the open VLMs tested here do not yet show those signatures. These parallels are computational. In the changing bandit, participants' estimated forgetting increased with hazard rate, while model estimates varied less even when told the rate; the paired differences were not significant after correction. The two tasks probe complementary demands of adaptive inference, extracting source-relevant evidence and discounting stale information, and we do not assume that they share one mechanism.

Limitations. The task simplifies Behrens et al. (2008): two perfectly opposed advisors, no separate reward stream, and no human data on this variant. It is binary, with fixed outcome quotas, which allow shortcuts and make past outcome labels informative about future ones, information that the correctness-only history removes; we therefore repeat the primary test on episodes with independent outcomes, where the gap was about half as large ($+.036$ to $+.083$) but significant in every model (Appendix I). The advice task did not reliably detect volatility-dependent updating in the Bayesian reference. We test this only in the changing bandit, where the human comparison is descriptive: participants were told the hazard rate and had played earlier games, and the test has about half the power for an effect of the human size. The summary discards trial order, and the size of the summary–raw gap depends on the history's layout (Appendix I). The open models (up to 38B, four of five built on Qwen3) were prompted to answer directly; the proprietary models were tested on a subset and differ in several respects at once. In the interpretable advice-task comparisons, images appear only as current reports. The analysis plan was fixed by a timestamped commit; deviations are listed in Appendix J.

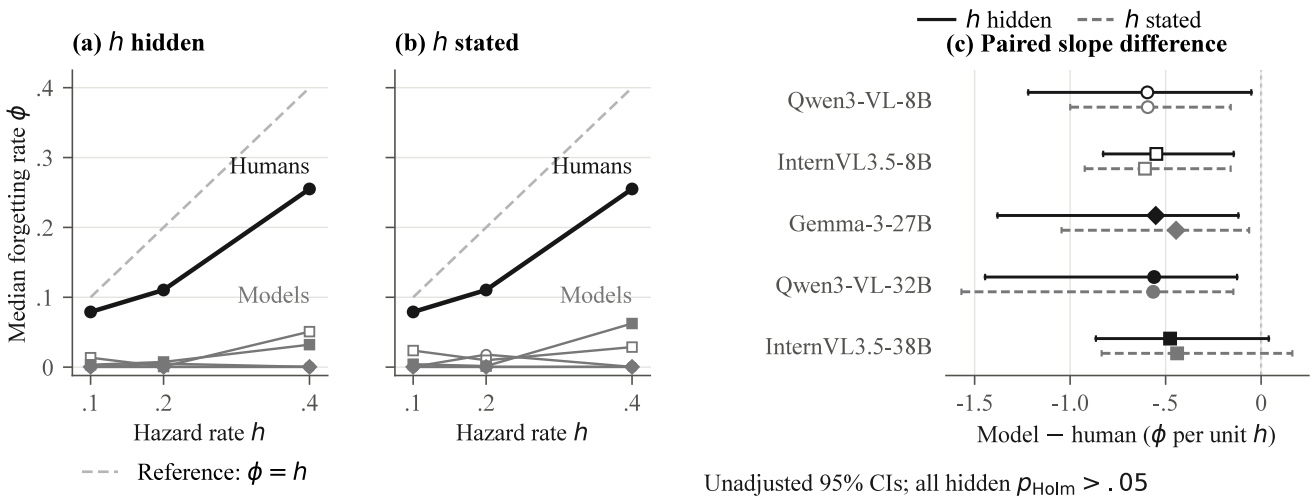


Figure 5: Comparisons between humans and the five open models remain descriptive. (a,b) Median fitted forgetting rate ϕ by hazard rate, with h hidden or stated for the model; humans were told h in both panels. Model estimates are medians over 20 fits to sampled binary-choice sequences. (c) Paired model-minus-human slope differences for the same 30 participant histories, with unadjusted 95% CIs from resampling participants. Solid and dashed intervals mark hidden and stated h; marker shape and fill identify the five models. None of the five hidden-h comparisons is significant after Holm correction. The simulation study shows limited power and an upward bias (Appendix H); these results do not establish equivalence.

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Supplementary Material

The appendices provide exact task inputs, analysis definitions and checks of alternative explanations. Development-only checks are marked separately from held-out evidence; pending analyses are not presented as results.

A. Prompts and Cover Stories

Every prompt has four parts: instructions naming the two advisors and the cover story, the history in one of the formats of the main text, the current reports, and the question.

Prompts are passed through each model's chat template as a single user message. For InternVL3.5 the template adds the model's default system message (in Chinese) and the suffix "Answer the question using a single word or phrase." All answer words are single tokens in every tokenizer used.

Task details. Each episode is shown with two cover stories that share the same outcome sequence: a pet cover story, in which reporters say whether a hidden animal is a cat or a dog, and one of six other cover stories (colors, weather forecasts, stock predictions, sports predictions, medical readings, news claims), assigned in blocks so that cover story is crossed with phase order. Cover stories change only the wording, the answer words and the report images; all outcome sequences are synthetic. Current reports are text or an image: a programmatically drawn icon without text for the non-pet cover stories, and a photograph from the Oxford-IIIT Pet dataset (Parkhi et al. 2012) for the pet cover story (with text reports the pet cover story is text only, like the others). Both advisors' correctness is revealed on every trial (full feedback). In the counterbalanced copy, advisor names, answer labels and the order in which the instructions introduce the advisors are swapped while outcomes and images are identical; within each trial the two reports are listed in random order. The instructions say that an advisor's accuracy may change over time but give no rate. The held-out set has eight episodes per non-pet cover story.

Table 3 lists the cover stories. The examples below are rendered by the experiment code from the fixed episode files; line breaks are as in the prompt and *[image]* marks where an image is inserted.

Cover story	Advisors	Answers	Report image
color	reporters	red/blue	color patch
weather	forecasters	rain/dry	cloud, sun
stock	analysts	up/down	trend arrow
sports	commentators	win/lose	happy/sad face
medical	doctors	positive/negative	plus/minus
news	news sources	true/false	tick, cross
pet	reporters	cat/dog	photograph

Table 3: Cover stories with the selected icon set (v3.1). Trials are called trials (color, pet), days, weeks, games, cases or stories. Icons are drawn by code and contain no answer words. Appendix G shows the original and selected medical and sports icons and identifies results obtained with the original set.

Example 1: raw history, weather, held-out episode, trial 6.

> You are observing a sequence of days. On every day, two weather forecasters, Tim and Joe, give conflicting forecasts for the day's weather (rain or dry). After each day, the actual weather is observed. A forecaster's accuracy may change over time.
> Past days:
> Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 2: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Day 3: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Day 4: Joe forecasts rain; Tim forecasts dry. Actual weather: dry.
> Day 5: Tim forecasts rain; Joe forecasts dry. Actual weather: rain.
> Current day 6 (actual weather not yet observed):
> Joe forecasts: dry
> Tim forecasts: rain
> Predict the actual weather on the current day. Answer with exactly one lowercase word: rain or dry.

The other formats change only the history lines. In the raw history with correctness, day 1 reads "Day 1: Joe forecasts rain; Tim forecasts dry. Actual weather: dry. (Joe was wrong; Tim was correct.)". In the correctness-only history it reads "Day 1: Joe was wrong; Tim was correct.". In the summarized history the five history lines and the "Past days:" line are replaced by "Summary of all 5 past days: Joe was correct 0 times; Tim was correct 5 times.". In the correctness chart they are replaced by "Past days 1-5 are summarized in this chart (days run left to right in four lines of 20; for each day, a green square means that forecaster was correct, a gray square means that forecaster was wrong, an empty square means not yet played):" followed by the chart (448 px; one row per advisor in each line). In the judgment readout the current reports and question are replaced by "Before day 6: which forecaster is currently more likely to be correct? Answer with exactly one name: Joe or Tim.".

Example 2: described accuracy, weather, development episode. The described-accuracy control replaces the history with the following passage (advisor names vary by item); the remaining instruction and query structure is unchanged:

> Summary of past performance: Jim was correct on 80% of past days; Ken was correct on 20% of past days.

The described-counts control replaces the summary line with "Summary of the last 10 days: Jim was correct 8 times; Ken was correct 2 times.".

Example 3: pet cover story with image reports, held-out episode, trial 4.

> You are observing a sequence of trials. On every trial, two reporters, Jon and Jim, give conflicting reports about a hidden animal. After each trial, the true animal is revealed. A reporter's accuracy may change over time.
> Past trials:
> Trial 1: Jon reports cat; Jim reports dog. True animal: cat.
> Trial 2: Jon reports cat; Jim reports dog. True animal: cat.
> Trial 3: Jim reports dog; Jon reports cat. True animal: cat.
> Current trial 4 (true animal not yet revealed):
> Jim reports: *[image]*
> Jon reports: *[image]*
> Predict the true animal on the current trial. Answer with exactly one lowercase word: cat or dog.

With text reports the images are replaced by the answer word.

Binding variants. The binding controls change only how the current reports of Example 3 are attached to the advisors (described accuracy, image reports). *Name on image*: the lines are as in Example 3, but each image carries its advisor's

name printed in a banner above the picture. *Composite image*: the two report lines are replaced by "Reports (one panel per reporter, name above each panel): *[image]*", one image with two panels, each headed by an advisor's name. *Name after image*: each report is given as "*[image]* (the image above was reported by Jim)", so the attribution follows the image. The composite-read check shows only the composite image and asks "The image has two panels, each with a name above it. What animal is in Jim's panel?".

B. Episode Generation and Counterbalancing

Schedules. Each episode has 80 trials in eight blocks of 10. Advisor A is the advisor favored on trial 1. Stable-first episodes have a stable phase (A correct with probability 0.8) followed by a volatile phase with blocks at 0.2, 0.8, 0.2, 0.8; volatile-first episodes have a volatile phase with blocks at 0.8, 0.2, 0.8, 0.2 followed by a stable phase at 0.8. Every episode therefore has four reversals (trials 41, 51, 61, 71 in stable-first and 11, 21, 31, 41 in volatile-first episodes). Advisor B is always correct when A is wrong. For comparison, the advice schedule of Behrens et al. (2008) (Supplementary Information) has 120 trials: 75% correct on trials 1–30, alternation between 80% correct and 80% incorrect every 10 trials on trials 31–70, and 85% incorrect on trials 71–120. Our volatile phase matches their alternation; our stable phase does not.

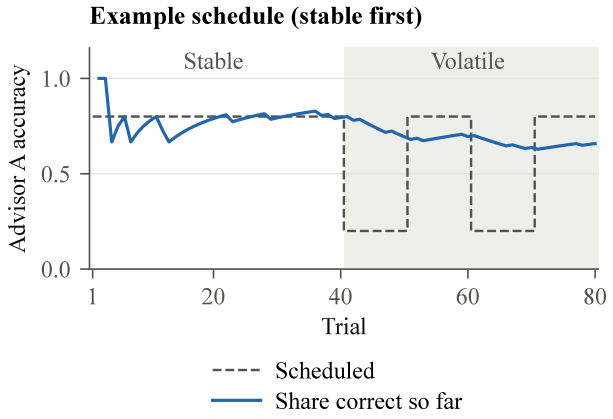


Figure 6: Why all-trial counts lag after reversals. Scheduled accuracy of advisor A and its cumulative observed accuracy before each trial in one stable-first development episode. Shading marks the volatile phase; the cumulative count retains evidence from earlier blocks.

Relation to the original human study. Table 4 identifies the simplifications of Behrens et al. (2008). Their participants learned both reward-based option values and the accuracy of a confederate's advice. We retain changing advisor accuracy but omit the separate reward stream, reward magnitudes and in-person confederate; the schedules also differ. This is an adaptation of the social-learning component, not a replication of the human experiment. No human data exist for our variant, so the Bayesian learner is a computational reference, not human performance; Psych-101 provides the separate human comparison (Appendix H).

Stratification. Within each block, A is correct on exactly $10p$ trials (8 or 2) in random order, and each outcome label is the truth on exactly five trials. This keeps realized accuracy equal to the schedule in every episode. It also fixes outcome quotas, which allow a shortcut that ignores the advisors: predicting the label with more occurrences remaining in the current block's quota (which presupposes knowing the quota) reaches 0.661 accuracy on the development episodes. We therefore also use a robustness set (below).

Names, order and counterbalancing. Advisor names are drawn from 16 common English first names that are single tokens in all tokenizers. The order in which the two reports are listed is random on each trial. Phase order alternates across episodes, the order in which the instructions introduce the two names alternates in pairs, and non-pet cover stories rotate in blocks of four episodes, so cover story, phase order and introduction order are fully crossed. Each episode has a counterbalanced copy with the advisor names swapped (A keeps its role), the answer-label mapping reversed and the introduction order reversed; outcomes and images are identical. Analyses average the label-aligned logits within each pair.

Splits and seeds. The main set has 16 development episodes (color, weather, stock and sports cover stories) and 48 held-out episodes (eight per non-pet cover story), 128 with counterbalanced copies, generated with seed 20261111. Pet photos come from the Oxford-IIIT Pet dataset (CC BY-SA 4.0): 128 photos from the official trainval split for development episodes and 256 from the test split for held-out episodes, with no overlap. The robustness set has 24 held-out episodes (48 with copies; seed 20261112) in which correctness is drawn independently on each trial, outcome labels are drawn independently, and volatile-phase dwell times are drawn uniformly from 6–16 trials. All data are regenerated from seeds and checked against stored SHA-256 hashes.

C. Statistical Analysis, Control Conditions and Decomposition

Statistical analysis. The unit of analysis is the episode pair; all measures are averaged within an episode and its counterbalanced copy first, after averaging the label-aligned logits, which cancels additive name and label biases. Besides accuracy we report the probability of following the currently more accurate advisor by phase, and learning curves aligned to reversals. The primary test, fixed before held-out evaluation (timestamped commit; not a formal preregistration), is the summarized minus raw history accuracy for text reports, per model, with a two-sided Wilcoxon signed-rank test and Holm correction across models; the decision criterion is a gap significantly above zero in both language-model families. Raw-history accuracy is called near chance only within 0.5 ± 0.03. Secondary tests are two one-sided tests (TOST; Lakens 2017) of whether a model

	Behrens et al. (2008)	This study
Agents	24 human participants (fMRI)	Five VLMs without fine-tuning
Choice	Blue or green option, each with a random reward magnitude	Which of two binary reports is correct (whose advice to follow)
Sources of information	Reward history of the options and one confederate's advice	Two advisors whose reports always conflict
Feedback	Correct option revealed after each trial	Outcome revealed after each trial
Advice schedule	120 trials; 75% correct, then 80%/20% alternating every 10 trials, then 15%	80 trials; 80% (stable, 40 trials) and 80%/20% alternating every 10 trials (volatile, 40 trials), both orders
Behrens-inspired Bayesian reference on our episodes (trial-by-trial advisor correctness, no VLM)		
Accuracy	.672 (development), .668 [.661, .674] (held-out), .660 (robustness); all-trial count rule: .630 / .632 / .594	

Table 4: The original human study and ours. The original advice schedule shows one counterbalanced orientation from the Supplementary Information of Behrens et al. (2008); its reward schedule is omitted. fMRI: functional magnetic resonance imaging. Accuracy of the Bayesian learner: proportion of trials ($t \geq 2$) on which it favours the advisor who is correct, with a 95% bootstrap CI over episodes.

given a summary matches the count rule alone, with equivalence bounds of ± 2 percentage points; the position of the correctness-only history between raw and summarized history, $(\mathrm{acc}_{\text{corr. only}} - \mathrm{acc}_{\text{raw}})/(\mathrm{acc}_{\text{summary}} - \mathrm{acc}_{\text{raw}})$, interpreted only when the summary–raw gap exceeds 0.05 and its CI excludes zero; and the primary test on the robustness set, as a separate family. CIs are 95% bootstrap intervals over episode pairs (10,000 resamples). We report per-story estimates and a sign test across the six non-pet cover stories (smallest attainable two-sided $p = .031$). Development episodes are used for exploratory checks and setting choices such as the window length K; held-out episodes are run once. The kernel regression uses lags $k = 1, \ldots, 15$ plus regressors for repeating the previous outcome, a label bias and listing order; strategies for κ are repeating the previous outcome, win-stay/lose-shift (WSLS) on advisors (following the advisor correct on the previous trial, as in Behrens et al. 2008), counts over all or recent trials, BOCPD and a label bias. These analyses describe behavior consistent with a strategy and do not identify a mechanism.

How much of the gap does correctness recover? On held-out episodes, correctness-only histories recovered 89–101% of the summary−raw accuracy gap (Table 7). The ratio's uncertainty matters because a near-complete recovery is not an equivalence claim. Table 1 reports the accuracy means and paired summary−raw differences; Table 7 supplies the individual accuracy CIs and the recovery ratio with its CI.

Control conditions. Table 6 gives each control condition with text and image reports, and Table 7 the fraction of the main-task gap recovered by correctness-only histories. Image reports lowered accuracy in almost every control condition, including described accuracy (.59–.84 instead of .92–1.00); the one exception was Qwen3-VL-8B with a long raw history (.533 vs. .531). The image-report results use the selected icons (v3.1, Appendix G). All selected icons were named correctly, and photo naming was .986–1.00, so single-image recognition alone does not explain these drops. Control conditions were run on development episodes for all five models.

D. Rule Simulations and Bayesian Reference

We ran two simulations without any VLM to check the design. Both score a rule by the proportion of trials ($t \geq 2$) on which the advisor it favours is correct; ties score 0.5.

Design simulation. The first simulation draws outcomes independently from the schedule (300 episodes per setting) and compares BOCPD (posterior predictive, $h \in \{1/8, 1/15, 1/30\}$), majority counts over the last $K \in \{3, 5, 7, 10, 15, 20, 30\}$ trials, counts over all past trials, and counts since the last true reversal. Across 20 settings (accuracy 0.65–0.85; phase lengths 40/40, 60/60 and 80/40; dwell 10 or 15 trials), the best BOCPD minus the best fixed window ranged from -0.004 to $+0.008$. For our setting (0.8; 40/40; dwell 10), accuracy was 0.687 for the best BOCPD ($h = 1/8$), 0.683 for $h = 1/15$, 0.690 for the best window ($K = 3$), 0.632 for counts over all trials and 0.739 for counts since the last reversal. Conclusions were the same for accuracies from 0.65 to 0.85, and accuracy varied by about one percentage point for hazard rates between 1/8 and 1/30.

Rules given to the VLM. Table 8 applies the actual summary rules and diagnostic references to our episodes. The last-five-trial count outperforms the change-point count on both sets; all-trial counts lag after reversals, and the ordering of the longer windows differs across sets. We therefore do not claim that a change-point rule is optimal or that these online rules are generally equivalent. The reversal-aligned reference uses reversals only up to $t - 1$, learning of each switch one trial late; it is not an upper bound. Knowing the current accuracy is likewise not a strict ceiling because the main schedule has quotas (Appendix B).

These rule baselines were computed on held-out episodes before any model results on the main data were seen. They are properties of the episode generator and involve no model behavior.

Bayesian reference implementation. Our hierarchical learner follows the volatility-learning idea of Behrens et al. (2007, 2008), using discrete Gaussian transition weights in logit accuracy; the original formulation uses a beta transi-

Model	Summary	Model − rule [90% CI]	$p_{\mathrm{TOST,Holm}}$	Choice agreement
Qwen3-VL-8B	All trials	−.011 [−.017, −.005]	.020	.941
	Change-point	−.012 [−.017, −.007]	.020	.970
	Last 5	−.024 [−.034, −.015]	.769$^{\times}$	.943
InternVL3.5-8B	All trials	+.003 [.000, .006]	<.001	.978
	Change-point	−.005 [−.007, −.003]	<.001	.985
	Last 5	−.011 [−.016, −.006]	<.001	.977
Gemma-3-27B	All trials	−.014 [−.019, −.010]	.053$^{\times}$	.935
	Change-point	−.011 [−.015, −.006]	.002	.965
	Last 5	−.015 [−.021, −.009]	.088$^{\times}$	.958
Qwen3-VL-32B	All trials	.000 [−.003, .002]	<.001	.994
	Change-point	−.003 [−.005, −.001]	<.001	.993
	Last 5	−.001 [−.003, .000]	<.001	.997
InternVL3.5-38B	All trials	+.001 [−.001, .003]	<.001	.999
	Change-point	−.001 [−.001, .000]	<.001	1.00
	Last 5	.000 [−.001, .000]	<.001	1.00

Table 5: Secondary equivalence tests and agreement with the summary's rule on held-out trials. Differences and CIs are proportions. TOST uses the fixed ±2-percentage-point tolerance, with Holm correction over the three summaries per model; $^{\times}$marks failure to establish equivalence. Last 5 was selected on development episodes for every model. Agreement is the proportion of non-tied trials with the same choice as the rule; similar accuracy alone would not establish this. Primary summary−raw tests appear in Table 1; all primary Holm-corrected $p < 10^{-8}$.

Condition	Reports	Qwen3-8B	IVL-8B	Qwen3-32B	Gemma-27B	IVL-38B
Described %	Text	.920	1.00	1.00	.982	1.00
	Image	.844	.799	.683	.650	.594
Described counts	Text	.998	1.00	1.00	.978	1.00
	Image	.866	.786	.645	.652	.527
Short raw	Text	.703	.632	.607	.600	.786
	Image	.641	.542	.518	.592	.540
Short corr. only	Text	1.00	.996	1.00	1.00	1.00
	Image	.949	.743	.701	.674	.603
Long raw	Text	.531	.567	.556	.545	.696
	Image	.533	.520	.424	.516	.462
Long corr. only	Text	.676	.674	.759	.859	.900
	Image	.585	.449	.493	.598	.536
Single trial	Text	1.00	1.00	1.00	1.00	1.00
Described %, name on image	Image	.833	.833	.761	.719	.629
Described %, composite image	Image	.750	.853	.766	.810	.969
Described %, name after image	Image	.616	.699	.551	.536	.665
Composite read	Image	1.00	1.00	1.00	1.00	1.00
Chart read	Chart	.620	.688	.672	.609	.708

Table 6: Control conditions on development episodes (accuracy on control items, averaged over counterbalanced copies and pooled over cover stories). Lower block: binding variants of the described-accuracy condition with image reports (advisor name printed on the image; both reports in one composite image; name given after the image), reading one panel of the composite image, and reading one cell of the correctness chart. Qwen3: Qwen3-VL; IVL: InternVL3.5. Image reports use the selected icons (v3.1). Results by cover story will be included in the released analysis output.

tion in accuracy. The grid has 60 equally spaced accuracy values $p \in [.01, .99]$, 20 equally spaced log-volatility values $v \in [\log(.02), \log(3)]$, and a static, unknown meta-volatility $k \in \{.05, .2, .5, 1\}$, with a uniform joint prior. Before each trial, probability mass first moves across v with weights proportional to $\exp[-(v' - v)^2/(2k^2)]$, then across p with weights proportional to $\exp[-(\mathrm{logit}\, p' - \mathrm{logit}\, p)^2/(2e^{2v'})]$; each transition is normalized over destination grid points. The predictive mean $q_t = E[p_t \mid o_{1:t-1}]$ determines the choice before observing correctness o_t; a Bernoulli likelihood then updates the joint posterior, integrating over v and k. Accuracy follows A when $q_t > .5$ and B when $q_t < .5$, with half credit at ties. The implied learning rate is $(q_{t+1} - q_t)/(o_t - q_t)$, excluding the last trial and prediction errors below 10^{-9} in magnitude. This finite-grid reference is not fitted to VLM choices.

	Accuracy 95% CI (%)				Fraction recovered
Model	Raw	Raw+corr.	Corr. only	Summary	[95% CI]
Qwen3-VL-8B	[47.3, 50.0]	[50.3, 53.2]	[60.6, 62.9]	[61.2, 62.9]	0.98 [0.90, 1.07]
InternVL3.5-8B	[46.8, 48.9]	[49.4, 52.1]	[62.8, 64.2]	[63.0, 63.8]	1.01 [0.96, 1.05]
Gemma-3-27B	[45.6, 48.3]	[48.4, 51.3]	[59.0, 61.2]	[61.0, 62.5]	0.89 [0.83, 0.96]
Qwen3-VL-32B	[45.4, 48.2]	[50.9, 53.6]	[62.1, 63.5]	[62.7, 63.6]	0.98 [0.93, 1.03]
InternVL3.5-38B	[53.8, 56.0]	[54.9, 57.0]	[61.7, 63.2]	[62.9, 63.7]	0.90 [0.81, 1.01]

Table 7: Uncertainty in the full-set accuracies and correctness-only recovery. The first four columns give pointwise 95% episode-pair bootstrap CIs in percent for the accuracies in Table 1, with counterbalanced logits combined before scoring. The last column is the dimensionless fraction of the summary−raw gap recovered by correctness-only histories: $(\mathrm{acc}_{\mathrm{corr.\,only}} - \mathrm{acc}_{\mathrm{raw}})/(\mathrm{acc}_{\mathrm{summary}} - \mathrm{acc}_{\mathrm{raw}})$. Recovery CIs are reported only when the gap exceeds 0.05 and its CI excludes zero, as fixed in the analysis plan.

Rule or reference (no model)	Held-out	Robustness
All-trial count	.632	.594
Last 5 trials	.692	.675
Last 10 trials	.612	.635
Last 20 trials	.579	.604
Change-point count	.678	.663
BOCPD posterior observer	.677	.675
Since true reversal	.744	.735
Always follow A	.648	.547
Known current accuracy	.800	.788
Outcome quota	.655	–

Table 8: Rules on 48 held-out and 24 robustness episodes. Change-point count uses the inferred run's majority; the posterior observer uses BOCPD's predictive probability. Rows below the divider use hidden generator information and are references, not available model strategies. The robustness set has no outcome quota. Held-out 95% bootstrap CIs (2,000 episode resamples): last 5 [.685, .698], change-point count [.666, .688], all-trial count [.628, .636].

E. Change-Point Controller and Summary Variants

Change-point detection. Let $o_t = 1$ if advisor A was correct on trial t. Before trial t, the controller runs Bayesian online change-point detection (Adams and MacKay 2007) on $o_1, \ldots, o_{t-1}$ with a Bernoulli likelihood, a Beta(1,1) prior for each run and a constant hazard rate $h = 1/15$. This rate lies between the reversal rates of the volatile (1/10) and stable phases and was not tuned (Appendix D). The controller takes the maximum a posteriori run length r (the inferred number of trials since the last change), sets it to at least 3, and summarizes trials $\max(1, t-r)$ to $t-1$.

Summary variants. All variants use the same sentence as the summarized history, restricted to a range of trials ("Summary of the last n days (days s–$(t-1)$): Ken was correct x times; Sam was correct y times."). The variants are: all past trials (the primary summary); the last $K \in \{5, 10, 20\}$ trials; the run since the last inferred change-point (BOCPD); and the trials since the last true reversal up to $t-1$ (reversal-aligned reference). Two further conditions truncate the raw history instead of summarizing it: to the BOCPD run, or to the last 10 trials. The window length used in the secondary tests is, for each model, the one with the highest accuracy on development episodes, fixed before held-out evaluation; Table 9 reports the additional variants, and Table 5 gives rule agreement and equivalence tests. The last five trials were selected for every model. On held-out episodes, accuracy followed the rule in every variant: last 5 trials .667–.691 (rule alone .692), change point .666–.677 (.678), last 10 trials .605–.616 (.612), last 20 trials .581–.586 (.579). The model thus adds little to, and takes little from, the rule it is given; the choice of rule decides accuracy. The same holds for the reversal-aligned reference (.727–.744; rule alone .744). Truncating the raw history did not help: with only the last 10 raw trials, accuracy was .391–.467, at or below the full raw history (.468–.549) in every model, and truncating to the change-point run gave .435–.502 across all five models. Shortening this raw format did not recover the benefit of stating who was correct.

What the controller is. The controller is a training-free rule outside the model; it sees only the sequence indicating whether advisor A was correct and does not read model states. It is motivated by change-point models of human learning (Nassar et al. 2010, 2012), but we use it as an integration rule, not as a model of the VLM or of the brain.

F. Behavioral Diagnostics and Identifiability

What strategy agreement can establish. Cohen's κ adjusts agreement for chance. A rule that nearly always chooses one advisor can have high raw agreement and low κ; matching a strategy does not identify an internal algorithm. Figure 9 shows all raw-history strategy comparisons, and Fig. 8 retains each model's kernels and CIs rather than only the mean in Fig. 3. Table 10 checks these diagnostics on synthetic agents: even a constant-rate delta learner can agree most with BOCPD, while its kernel has the expected geometric decay.

Pending model comparisons. Episode-grouped cross-validation fits remain pending and support no model-selection claim in this draft. Both cover-story versions of an episode must be held out together; fits that hold them out separately are not used. The reported strategy and kernel

	Summarized history					Truncated raw	
Model	Last 5	Last 10	Last 20	Change-point	Since reversal	Last 10	Change-point
Qwen3-VL-8B	.667	.605	.582	.666	.727	.458	.477
InternVL3.5-8B	.681	.613	.586	.673	.737	.467	.472
Gemma-3-27B	.677	.609	.581	.667	.728	.391	.435
Qwen3-VL-32B	.691	.615	.582	.675	.740	.429	.453
InternVL3.5-38B	.691	.616	.581	.677	.744	.450	.502
Rule alone	.692	.612	.579	.678	.744	–	–

Table 9: Additional summary variants and truncated raw histories on held-out episodes; all-trial summaries appear in Table 1. The last-five-trial window was selected on development episodes for every model. Since reversal uses true change times hidden from the models and is a reference. Truncated raw histories have no separate rule; change-point truncation uses the same inferred run as the corresponding summary.

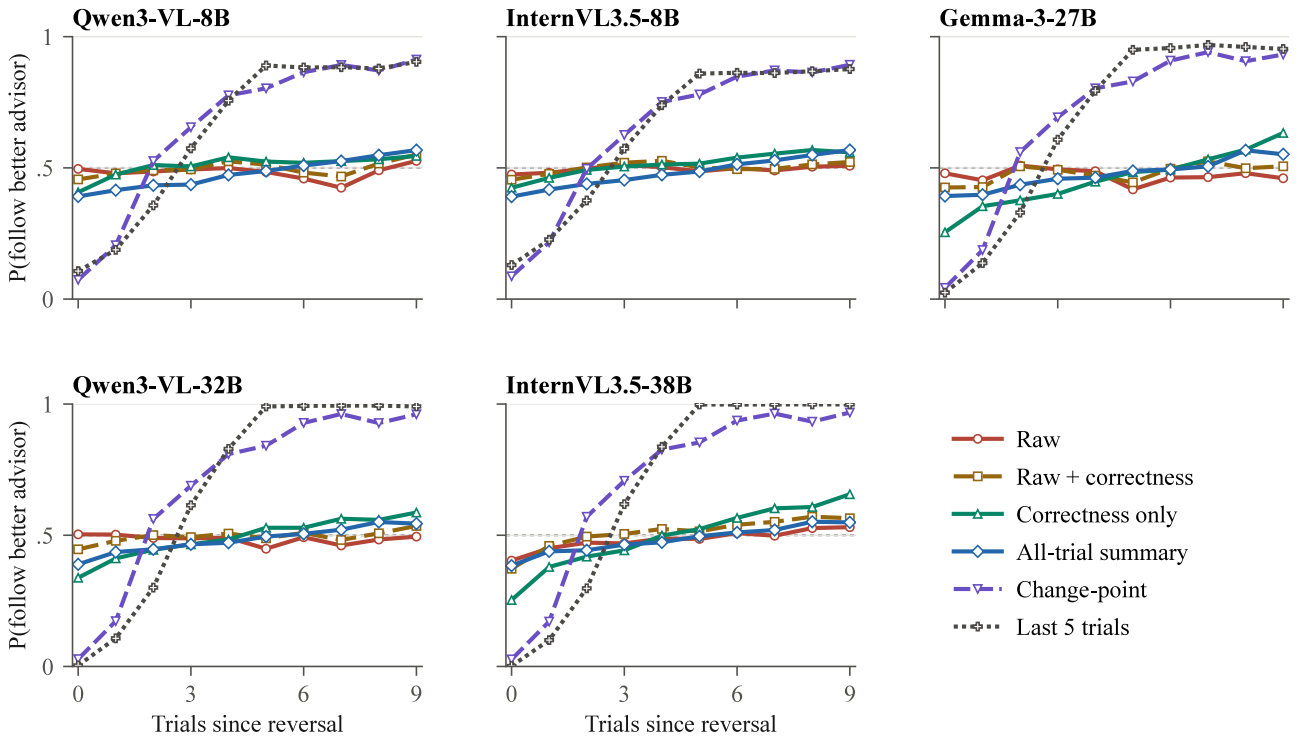


Figure 7: Following the currently better advisor after a reversal, separately for each open model and history format. Zero marks the first trial after a reversal. Each curve is a mean choice probability on held-out episodes; the panels are descriptive and do not show confidence intervals. Fig. 4 summarizes the same curves over the five fixed models.

analyses do not depend on these fits.

Constant-rate reference. Let $o_t = 1$ when advisor A is correct. The delta learner updates $q_{t+1} = q_t + \alpha(o_t - q_t)$ from $q_1 = 0.5$, with learning rate α. Its choice probability is $P_t = \sigma[\beta(q_t - 0.5) + b]$, where σ is the logistic function, β the inverse temperature and b a bias. This defines the synthetic delta agent and the constant-rate null below.

Effective learning rate. The update of the model's choice probability in logit space after each outcome, regressed on the signed outcome separately for the stable and volatile phases, measures how strongly the model responds to new outcomes. This statistic is not zero for a learner with a single constant learning rate: preferences are more extreme in the stable phase, where the logit stretches updates. We therefore calibrate it against a null in which a constant-rate delta rule fitted to the same episode generates simulated choice probabilities with the same residual noise (200 simulations); the calibrated volatile-minus-stable difference and its z score against the null indicate whether a model updates more in the volatile phase than a constant-rate learner would.

Recovery. Simulations with constant-rate, phase-specific and confirmation-dependent delta learners (100 each; logit

noise s.d. 0.3) recovered learning rates well but did not reliably identify the generating model: phase-specific learners were identified in only 40% of cases, and confirmation-dependent learners in 10%. Thus parameter recovery alone does not justify an algorithmic interpretation of the VLM's behavior.

Can these episodes reveal volatility-dependent learning? We checked this with our Bayesian reference learner, which allows volatility-dependent updating. On our 80-trial episodes its implied learning rate rose only from .100 in the stable to .115 in the volatile phase (held-out episodes). Neither the calibrated volatile-minus-stable effective learning rate ($+0.002\,[-0.010, 0.014]$ over 16 development episodes) nor phase-specific integration kernels (volatile/stable ratio of the summed weight of the last three outcomes: 0.84–0.90, versus 0.96 for a constant-rate learner) detected this change, so we do not use the advice task to test whether models adapt their learning rate to volatility. The changing bandit (Appendix H), in which volatility differs across whole games, tests that question instead.

G. Icon Design, Multimodal Binding and Chart Legibility

The legibility and binding controls use development episodes only and are exploratory; the comparison of text and image reports on the main task uses held-out episodes. All use the selected icons (v3.1) described below. Every model named the selected icons correctly. Pooled image naming was .988 for InternVL3.5-8B (.986 for pet photos) and 1.000 for the other models, so recognition alone does not explain the remaining contextual costs (Table 11). With described accuracy, accuracy in four models was lowest when the advisor's name was given only after the image (.54–.70) and higher when the name was printed on the image (.72–.83) or both reports were shown in one composite image (.75–.85), although each panel of a composite image was read correctly on every item; InternVL3.5-38B did best with the composite image (.97) and worst with plain image reports (.59). With correctness-only histories, image reports lowered held-out accuracy from .602–.636 to .523–.552. Reading a single cell of the correctness chart was below the prespecified 80% chart-reading criterion for every model (.61–.71).

Icon design and examples. Icons replace the answer word with a picture in the image-report conditions. Figure 10 shows the original and selected pairs at 224 pixels; InternVL3.5 receives 448-pixel renderings from the same drawing functions. In the original medical pair, two red test-strip lines indicate positive and one indicates negative. The selected pair uses a dark plus or minus inside the same circular outline. In sports, the original gold trophy and red cross on a grey ball encode win and lose; the selected pair uses a yellow happy face and a grey-blue sad face. Labels and selection markers in the figure are reader annotations and are absent from the report images. These are synthetic symbols for the answer labels, not clinical scans or photographs of game outcomes. All other cover-story icons are unchanged.

Development-only selection. We tested five medical pairs and four sports pairs with Qwen3-VL-8B and InternVL3.5-8B before held-out evaluation. The single-image probe asks for each of the two labels under both answer orders (four prompts per pair and model). We required all four responses to be correct for both models, then selected the eligible pair with the highest mean accuracy over four image-report controls and both models. The controls use described accuracy, a short raw history, a short correctness-only history, and a long correctness-only history. Each downstream cell contains 64 items after averaging label-aligned logits across the counterbalanced episode copies. This selected plus/minus and happy/sad faces. Enlarging the test strip left naming accuracy at .50 for both models, so size alone did not resolve that failure. Coloured plus/minus signs scored higher downstream but failed the InternVL3.5 naming criterion.

Results and limits. The downstream mean increases from .521 to .668 for medical and from .588 to .678 for sports. Both selected pairs reach 4/4 in the naming probe for both models. However, accuracy in the described-accuracy condition, pooled over these two image-report cover stories is .773 for Qwen3-VL-8B and .727 for InternVL3.5-8B, below the 80% described-accuracy criterion. Single-image recognition therefore does not establish successful use of image reports in the advice-taking task. Selection and evaluation use the same development controls, so these gains are descriptive and do not estimate held-out improvement.

Versioning. Icon selection was a development-driven deviation from the fixed plan. The selected pairs define v3.1; original v3 outputs remain separate. Text reports, pet photographs, the correctness chart and the primary text-report comparison are unchanged. The three models not used for selection also named the selected icons correctly. Tables 6 and 11 use the completed v3.1 reruns; the development gains above are not held-out evidence of improvement.

H. Human Comparison: Protocol and Validity Checks

Task and data. In the changing bandit of Xiong et al. (2023), participants repeatedly choose between two slot machines whose payoffs stay constant until a change point, which occurs with a hazard rate h; payoffs are noise-free, so a normative learner forgets the value of the unchosen machine at rate $\phi = h$. We use the 30 participants in Psych-101 (Binz et al. 2025) and their three 100-trial games ($h = 0.1, 0.2, 0.4$; 90 games, 9,000 trials). Participants were told h, and each 100-trial game directly preceded a 1,500-trial game with the same h, so by their second and third 100-trial games participants had played 1,600 and 3,200 trials; models see each game on its own.

Replay. Only the five open models are evaluated in this task; the proprietary API models are not included. Models are conditioned on each participant's previously executed behavior (Binz et al. 2025): before trial t the model sees the original instructions and the participant's choices and payoffs on trials $1, \ldots, t-1$, and is asked which machine to press (one letter). The readout is the choice probability between

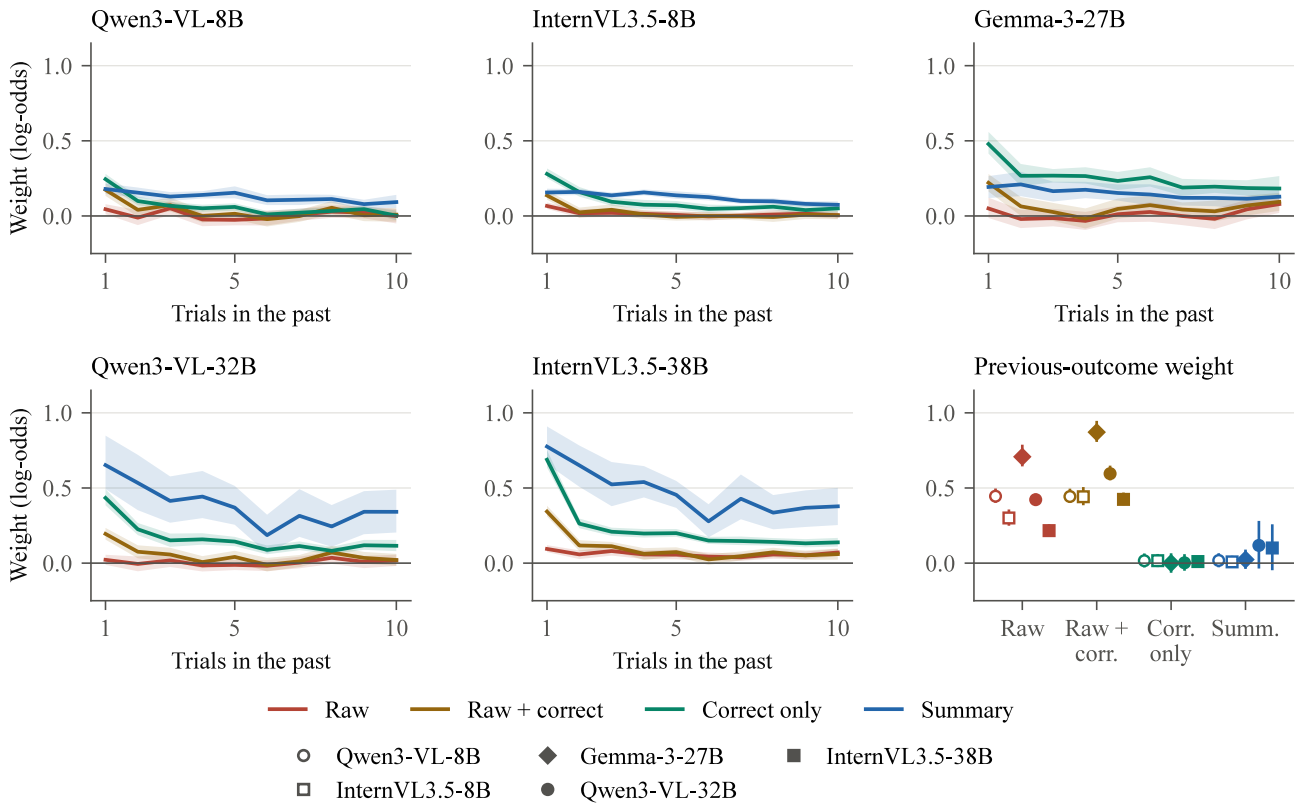


Figure 8: Integration kernels per model (shaded: 95% bootstrap CI over episodes) and, bottom-right, the weight on repeating the previous outcome by history format (bars: 95% CIs; held-out).

Synthetic agent	Best strategy	κ	Repeat w	Lag-1 w
Repeat last outcome ($w = 3$)	Repeat last outcome	1.00	3.00	0.00
Delta rule ($\alpha = 0.3$, $\beta = 8$)	BOCPD	0.94	0.00	1.20
50% copy, 50% delta	Repeat last outcome	0.49	1.21	0.56
Last-5 counts	Last 5	1.00	0.02	1.08
All-trial counts	All-trial counts	1.00	0.05	0.27
Label bias ($w = 3$)	Label bias	1.00	0.00	0.00

Table 10: Strategy and kernel analyses applied to synthetic agents on the main episodes (development and held-out; no VLM). Best strategy: highest Cohen's κ; Repeat w and Lag-1 w: kernel regression weights on repeating the previous outcome and on the advisor correct one trial back. For the delta rule the fitted kernel decays geometrically with ratio 0.700 (true $1 - \alpha = 0.7$); the last-5 agent has a flat kernel up to lag 5. Strategies are correlated on this schedule, so the delta rule agrees best with BOCPD; its fitted kernel nevertheless recovers the expected geometric decay. Results on the robustness set were the same.

the two letters from the first-token logits, as in the main task. In the primary condition the hazard rate is removed from the instructions (hidden); in a secondary condition the original instructions state it (told). Model and participant thus share the within-game history but not the hazard information (hidden) or the prior experience (both conditions).

Fits. We fit an adapted myopic model-free learner based on Xiong et al. (2023), separately for each hazard rate. It has free rates for updating the chosen machine's value and forgetting the unchosen one toward the midpoint (ϕ), plus choice sensitivity, perseveration and a bias. Humans give binary choices, so for comparability we draw 20 binary choice sequences from the model's choice probabilities in each game, fit each, and take the median. The primary statistic is each participant's slope of ϕ on h; the comparison is the paired difference between the model's and the human's slope (Wilcoxon test, participant-bootstrap 95% CI), with hidden and told conditions shown in Fig. 5. The human fits are our re-analysis; Xiong et al. (2023) report no learning rates by hazard rate. In our re-analysis, the median human ϕ rose from .079 ($h = 0.1$) to .110 and .255 ($h = 0.4$; per-participant slope 0.47 per unit

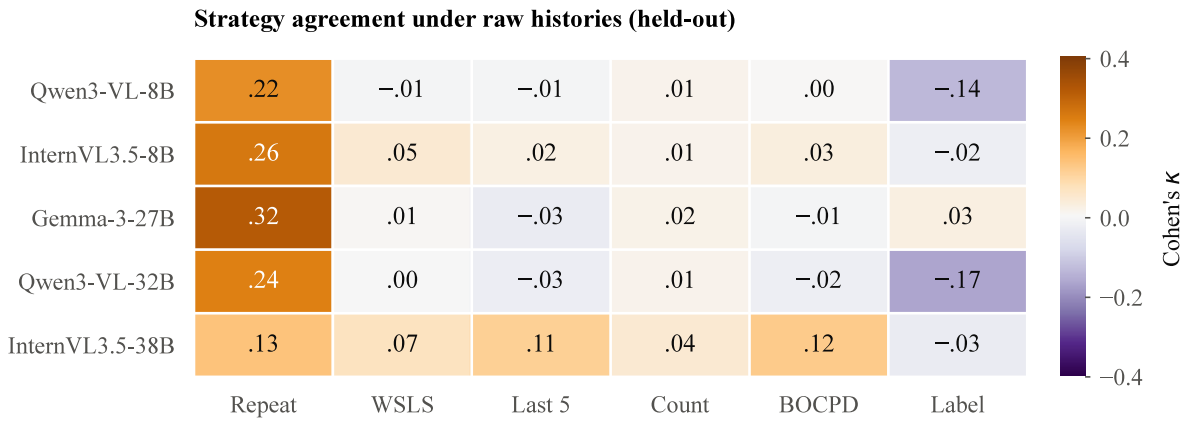


Figure 9: Under raw histories, repeating the previous outcome agrees most with the open models. Cohen's κ on held-out episodes, with a colour scale centred at zero. Repeat: repeat the previous outcome; WSLS (win-stay/lose-shift): follow the advisor correct on the previous trial; Last 5 and Count: recent and all-trial correctness counts; BOCPD: change-point rule; Label: outcome-label bias. Values are rounded to two decimal places; rounding zeros has no effect on their colour.

	Image	Main-task reports	
Model	naming	Text	Image
Qwen3-VL-8B	1.000	.619	.552
InternVL3.5-8B	.988	.636	.544
Gemma-3-27B	1.000	.602	.537
Qwen3-VL-32B	1.000	.628	.527
InternVL3.5-38B	1.000	.625	.523

Table 11: Image recognition versus use in context. Naming uses development items (photos and selected icons); main-task accuracy uses held-out episodes with correctness-only history. Composite-panel and chart-reading controls appear once, in Table 6. Naming a report image does not establish successful use of that report in the task.

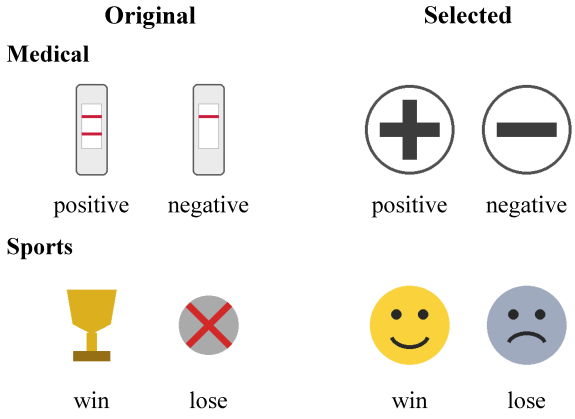


Figure 10: Original and selected report icons for medical (positive/negative) and sports (win/lose). Labels and headings are reader annotations and are absent from model inputs. Selection used development items only; the original recognition failures motivated the replacements.

h, positive in 73% of participants, Wilcoxon $p = .008$), adjusting for game position did not remove the increase (0.64 [0.23, 1.05]), and in the following 1,500-trial games the increase was weaker (median ϕ .076, .114, .131; slope test $p = .029$).

Calibration of the human–model comparison. We simulated the complete pipeline 200 times per condition. Simulated participants follow each real participant's fitted strategy at each hazard rate and play on the real value sequences; the simulated model follows the same strategy with a nominal forgetting-rate slope offset δ (rates clipped to $[.005, .95]$), and is replayed on the simulated participant's history; both are then fitted exactly as the real data (20 sampled choice sequences per model episode, median fit, paired Wilcoxon test and bootstrap CI). With $\delta = 0$, the test rejected at $p < .05$ in 6.0% of replications (Monte Carlo standard error 1.7%) and the 95% CI covered the true difference in 93.5%, but the estimated difference was biased upward by about $+0.09$. Power was 0.49 for $\delta = -0.5$ and 0.59 for $\delta = +0.5$ (realized median slope differences -0.50 and $+0.39$ after clipping), with CI coverage 0.87 and 0.96. We therefore treat the comparison as supporting and descriptive (main text).

Image and chart formats. With the hazard rate hidden, the changing bandit is also run with the last five payoffs shown as pictures of the machine's display instead of text (224 px; 448 px for InternVL3.5), and with the whole history drawn as one scatter chart of payoff by trial for each machine (448 px). Each has a perception check: reading the number on one machine picture (four options; 450 items), and reading

which machine was pressed, or which gave more points, at a given trial of the chart (540 items). These exploratory formats are interpreted only when their perception checks succeed. Every model read the number on a machine picture (.996–1.00), but none read the chart reliably (which machine was pressed: .56–.61; which gave more points: .50–.72), so chart results are not interpreted. With picture payoffs, the sampled-choice fits also gave low forgetting rates (median at most .10; per-participant slopes -0.16 to 0.07). None of the five slope tests reached an uncorrected $p < .05$. Gemma-3-27B had the largest positive slope (.073, $p = .060$), with a paired model-minus-human difference of -0.23 [-1.02, 0.21].

Model-free check of forgetting (exploratory). For every trial on which both machines had been observed, we regressed the choice of the first machine (participants: binary; models: choice probability) on the difference between the two machines' last observed values (per 10 points), separately when the older of the two values had been observed at most two trials earlier (fresh) or at least four trials earlier (stale), with the previous choice as a covariate (Table 12). All models were sensitive to value. We resampled the 30 participants as whole clusters (2,000 replicates), refitting the regression and using the same sampled participants for each human–model comparison. The stale-minus-fresh coefficient was negative for participants at $h = 0.1$ ($-.263$ [$-.501$, $-.114$]); its CIs included zero at the other hazards. Model-minus-human contrasts were positive at $h = 0.1$ and $h = 0.4$ in all five models (pointwise CIs above zero; Table 12). This supports a difference in stale-value use under replay, without establishing a hazard-dependent adaptation deficit.

I. Robustness, Prompt Format and Masking

Robustness set and cover stories. On the robustness set, where correctness and outcome labels are drawn independently on each trial and reversals are irregular, raw-history accuracy was .507–.520 in four models (near chance) and .559 in InternVL3.5-38B, and summaries reached .582–.595, close to the count rule on these episodes (.594). The summary$-$raw gap was smaller than on the main set but positive and significant in every model ($+.036$ to $+.083$, Holm-corrected $p < .001$; Table 13), so it does not depend on the outcome quotas. On held-out episodes the gap was positive in every non-pet cover story for every model (sign test $p = .031$, the smallest attainable) and in the pet cover story ($+.080$ to $+.179$; Table 14); it was smallest for Gemma-3-27B with the medical and news cover stories ($+.036$, $+.047$).

Which part of the raw history blocks the correctness statement. Appending who was correct to the raw history recovered only part of the summary–raw gap, and the weight on repeating the previous outcome stayed large. To find out why, we kept the layout and length of that history and replaced parts of each past trial with a dash ("Day 3: Ken forecasts —; Sam forecasts —. Actual weather: rain. (Ken was correct; Sam was wrong.)"): the outcome label, the two reports, or both. With both replaced the history carries the same information as the correctness-only history at nearly the length of the raw history with correctness. On development episodes, replacing the outcome label raised accuracy by 7.4 and 9.6 points (main effects; 95% CIs [5.2, 9.5] and [7.6, 11.5]) and lowered the weight on the previous outcome from .48–.50 to .12 (Table 15), whereas replacing the reports had no effect on its own ($+1.1$ and -0.2 points). With both replaced, accuracy was 1.5–2.1 points below the correctness-only history, leaving 14–20% of the original accuracy difference despite retaining much of the surrounding text; this comparison does not isolate additive contributions of length, layout and outcome labels. The outcome label thus elicits repeating the previous outcome even when who was correct is stated. Predictions for the held-out replication in five models were written before it was run. On held-out episodes, paired accuracy changes (Fig. 11) and repetition weights (Table 15) supported all four predictions in all five models. Replacing the outcome label raised accuracy over the raw history with correctness ($+2.7$ to $+6.4$ points; all $p < .001$) and cut the weight on the previous outcome from .42–.87 to .08–.23, whereas replacing the reports showed no clear accuracy change (between -0.4 and $+0.3$ points; all 95% CIs include zero) and left that weight at .26–.65. Averaged over the other factor, replacing the outcome label had a larger effect than replacing the reports in every model (difference of main effects $+2.9$ to $+6.8$ points, all CIs above zero). With both replaced, accuracy was $+7.3$ to $+10.7$ points above the raw history with correctness and within -4.0 to $+0.8$ points of the correctness-only history, despite retaining the longer layout. The held-out effect of replacing the outcome label alone is smaller than on development episodes and recovers 40–60% of the difference between the raw history with correctness and the correctness-only history.

Prompt wording (exploratory, development episodes). We re-ran the four text history formats for Qwen3-VL-8B and InternVL3.5-8B with two alternative wordings; episodes, names, answer words, list order and readout were unchanged. Reworded sentences keep the original layout with different phrasing ("In trial 3, Sam said cat and Ken said dog; it turned out cat."; "Tally over the 3 previous trials: Ken was right 2 times and Sam was right 1 times."). A compact record uses one line per trial ("#3 | Sam=cat | Ken=dog | truth=cat"; "Correct counts so far (3 trials): Ken=2, Sam=1."). Table 16 gives the results. With reworded sentences, every result was preserved: raw history at .47–.49, summary$-$raw $+.12$ and $+.15$, the correctness-only history at or above the summary, appending correctness recovering about 40% of the gap, and a positive weight on the previous outcome. The compact record changed the picture in both directions. Raw-history accuracy rose to .55–.56 and the weight on the previous outcome fell, whereas the correctness-only and summarized formats fell to .56–.60, and InternVL3.5-8B followed the count summary on only 72% of non-tied trials (98% with the original wording). The summary$-$raw gap shrank to $+.03$ [.01, .06] (Qwen3-VL-8B) and $+.01$ [$-.02$, .05] (InternVL3.5-8B). Raw-history accuracy stayed below the count rule (.63) in every wording, but the size of the raw–summarized gap depends on how the history is laid out, and we interpret it only for the natural-language layout used in all main conditions.

Completed compact-record extensions on H200 gave summary$-$raw gaps of $+.120$ [.097, .146] (Qwen3-VL-32B),

Agent (hazard hidden)	Stale − fresh value weight [95% CI] $h = 0.1$	$h = 0.2$	$h = 0.4$
Participants (told)	−.263 [−.501, −.114]	−.049 [−.170, .046]	−.040 [−.122, .034]
Qwen3-VL-8B	+.047 [−.037, .146]	−.009 [−.081, .080]	+.089 [.048, .133]
InternVL3.5-8B	+.100 [.052, .147]	+.057 [.016, .101]	+.087 [.054, .119]
Gemma-3-27B	+.049 [−.011, .110]	+.035 [−.016, .094]	+.053 [.007, .097]
Qwen3-VL-32B	+.040 [−.017, .102]	+.026 [−.047, .097]	+.099 [.052, .138]
InternVL3.5-38B	+.146 [.066, .229]	+.039 [−.020, .106]	+.105 [.070, .141]
	Model − human contrast in stale − fresh weight [95% CI]		
Qwen3-VL-8B	+.310 [.159, .552]	+.039 [−.063, .169]	+.129 [.049, .212]
InternVL3.5-8B	+.363 [.216, .598]	+.105 [.021, .217]	+.127 [.055, .208]
Gemma-3-27B	+.313 [.156, .541]	+.084 [−.007, .198]	+.093 [.014, .180]
Qwen3-VL-32B	+.303 [.139, .557]	+.075 [−.024, .189]	+.140 [.071, .214]
InternVL3.5-38B	+.410 [.236, .663]	+.087 [−.033, .225]	+.145 [.075, .226]

Table 12: Exploratory stale-value regression. Coefficients are log-odds per 10 points of last-observed value difference. Negative stale-minus-fresh values indicate discounting of stale values. Pointwise 95% CIs use 2,000 participant-cluster bootstrap refits; the same participants are resampled for paired model–human contrasts. These are descriptive, unadjusted intervals, separate from the primary hazard-slope tests.

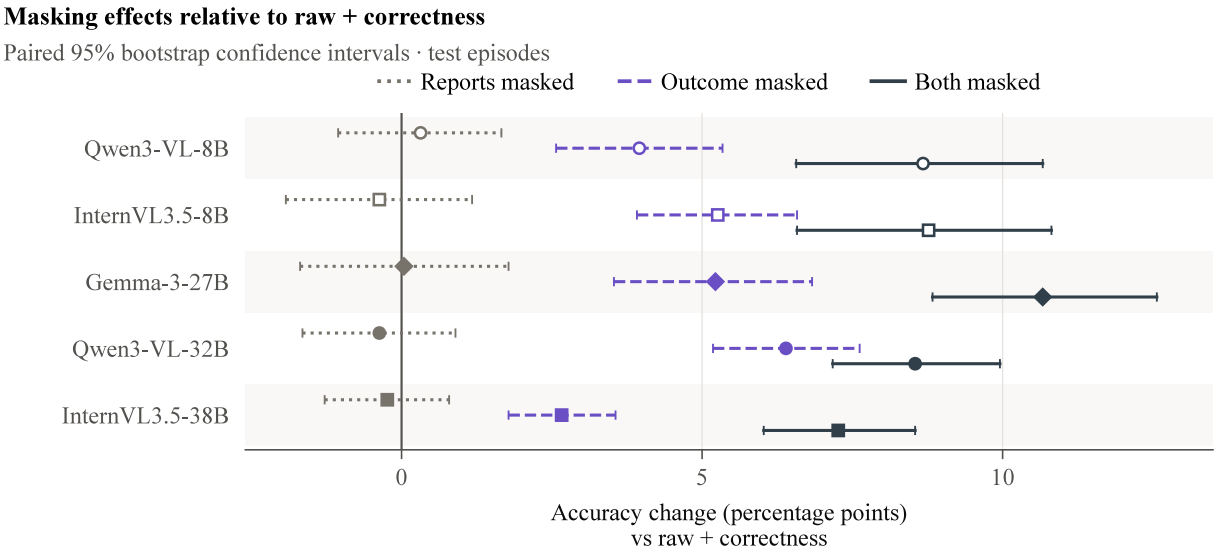


Figure 11: Paired masking effects corresponding to Table 2. Each contrast compares masking the reports, the outcome, or both against the raw history with correctness appended; correctness statements remain visible. Points show accuracy changes in percentage points, with pointwise 95% bootstrap CIs over the same 48 held-out episode pairs (10,000 resamples; text reports). The vertical line marks zero change. These are intervals for paired differences, not for the absolute accuracies in Table 2.

+.064 [.039, .087] (Gemma-3-27B) and +.071 [.051, .094] (InternVL3.5-38B), with paired 95% bootstrap CIs over the same 16 development episode pairs. Original-layout development runs are not all available on matched hardware, so these extensions do not isolate changes caused by layout.

The compact record changes the task description, history notation and current-trial question together; this sensitivity check does not isolate which change is responsible.

Judgment question and report position (exploratory). On raw-history trials the models were also asked which advisor was currently more likely to be correct (Appendix J). Table 17 compares the judgment averaged over counterbalanced copies with the choice on the same trials, and gives, for image and text reports with the correctness-only history, the share of choices for the first-listed report and accuracy by the position of the correct report. Within each trial the two reports are listed in random order, identical in both counterbalanced copies.

J. Models, Readouts and Analysis Transparency

Table 18 lists the Hugging Face checkpoints and pinned revisions.

Model	Raw	Summary	Gap [95% CI]	p
Qwen3-VL-8B	.507	.582	+.075 [.052, .100]	<.001
InternVL3.5-8B	.520	.594	+.074 [.042, .105]	<.001
Gemma-3-27B	.507	.590	+.083 [.055, .113]	<.001
Qwen3-VL-32B	.516	.593	+.077 [.051, .102]	<.001
InternVL3.5-38B	.559	.595	+.036 [.021, .051]	<.001

Table 13: Primary comparison on the robustness set (24 episode pairs with independent outcomes and irregular reversals). Accuracies are proportions; gap intervals are paired 95% bootstrap CIs over episode pairs. p: Wilcoxon signed-rank test, all Holm-corrected $p < .001$ within this family.

Readout. Models use bfloat16 with a float32 output projection. We extract first-token answer-word logits before processing and compute y_t as defined in Methods. Decoding is greedy; reasoning is disabled where the chat template permits. The raw-text judgment question asks "which forecaster is currently more likely to be correct?".

Readout quality. Table 21 reports full-vocabulary probability mass on the two answer tokens and greedy format following. For four models nearly all mass fell on the answer words in every format. Gemma-3-27B's share of trials with candidate mass below 0.5 was at most .03 for raw and correctness histories, but .40 for the all-trial summary (.59 with photo reports), where it often began a sentence ("Based on the summary. . . "). On these trials its single-copy accuracy was .51, versus .59 otherwise. Its summary accuracy and failure of equivalence with the count rule may therefore understate how well it would follow a summary in the requested format; this readout limitation also constrains its primary summary−raw comparison.

Sensitivity to counterbalancing the readout (exploratory). We additionally scored each counterbalanced copy separately, then averaged over copies and cover stories within each episode pair. All 48 held-out pairs and trials $t \geq 2$ were retained without candidate-mass filtering; ties received half credit. Every summary−raw gap remained positive (Holm-corrected $p < .001$; Table 19), although its size changed. This supports the format effect under per-copy scoring, still using candidate logits rather than unconstrained generation.

Proprietary models. GPT-6-Astra and Claude Fable 5.2 are queried through their APIs with the same prompts as the open models (text reports only), on a subset of held-out episodes and query trials: episodes 0–23 of the held-out set with their counterbalanced copies (four per non-pet cover story, half starting with each phase), every fourth trial ($t = 4, 8, \ldots, 80$), and the raw, correctness-only and all-trial summarized histories, 2,880 calls per model. Both were queried on 2026-11-13 with the default sampling settings (Claude Fable 5.2 does not accept a temperature), reasoning effort 'low' (the lowest that GPT-6-Astra accepts; Claude Fable 5.2 always reasons) and at most 4,000 output tokens; Claude Fable 5.2 was queried through the Message Batches API. The served model identifiers were gpt-6-astra and claude-fable-5-2 on every call. The choice is the answer word: either the whole answer has at most three words and contains exactly one of the two answer words, or its last line is one of them; all other answers are invalid. GPT-6-Astra answered with one word on every call (median 5 output tokens, including reasoning); Claude Fable 5.2 gave visible reasoning before a final-line answer on 47% of calls (77% with raw histories; median 47 output tokens), and 6 of its 2,880 answers (0.2%) were invalid.

Behaviour of the proprietary models (exploratory). On matched items with per-copy scoring, we tested advisor tracking against shortcuts using strategy agreement, learner–quota disagreement trials, genuine reversal windows, and a logistic regression of following A on its correctness at lags 1–10 and on repeating the previous outcome (Table 20). The outcome-quota rule uses the generator's hidden 5/5 outcome quota and is a diagnostic only. With raw histories both proprietary models gave large, decaying weights to past correctness (lag 1: 2.1 and 3.3 log-odds; lag 5: 0.7 and 1.4) and no reliable weight on the previous outcome; the open models' weights on past correctness were near zero and their weight on the previous outcome positive.

Image preprocessing. Icons and photos are rendered at 224 px for Qwen models and at 448 px (one tile, no tiling) for InternVL3.5. For Gemma-3, 224-px renders are resized by the processor to the fixed 896-px input (256 image tokens; pan-and-scan off). The correctness-only chart is rendered at 448 px for all models.

Software and hardware. Open-model held-out, robustness, Psych-101, control and window-selection evaluations used NVIDIA H200 GPUs (141 GB), one GPU per run, with PyTorch 2.8.0 (CUDA 12.8) and Transformers 4.57.1. The exploratory icon selection, prompt-wording checks for the two 8B models, and development masking checks used NVIDIA RTX 3090 GPUs with the same model revisions, precision and readout. The larger-model compact-record extensions used H200s with the held-out software stack. Each run stores the rendered prompt hash, the raw candidate logits, the probability mass on the two answer tokens, and a fingerprint of the configuration.

Code and data availability. Upon acceptance we will release the generator, seeds, prompts, runners and analyses. Scripts download Oxford-IIIT Pet (CC BY-SA 4.0) and the Psych-101 slice (Apache-2.0); third-party data are not redistributed. The original Xiong et al. (2023) data have no declared license and are used for analysis only. Generated episodes are verified by SHA-256.

Analysis plan and deviations. A timestamped commit fixed the plan before held-out evaluation; it was not formally preregistered. Prior observations comprised the 16-episode pilot, partial development controls from one model, rule-only held-out baselines, and unanalysed software-check outputs from a few held-out items. Before held-out model runs, the evaluation hardware changed and a 72B model and a 32B reasoning model were dropped for compute reasons. Development checks prompted icon replacements (Appendix G).

	Non-pet cover stories							
Model	Color	Weather	Stock	Sports	Medical	News	Sign test p	Pet
Qwen3-VL-8B	+.109	+.095	+.157	+.131	+.097	+.134	.031	+.148
InternVL3.5-8B	+.172	+.157	+.136	+.117	+.101	+.116	.031	+.179
Gemma-3-27B	+.160	+.146	+.171	+.157	+.036	+.047	.031	+.175
Qwen3-VL-32B	+.169	+.187	+.193	+.158	+.168	+.101	.031	+.164
InternVL3.5-38B	+.055	+.101	+.127	+.087	+.087	+.068	.031	+.080

Table 14: Summary−raw gap by cover story (held-out; eight episodes per non-pet cover story). Pet: the pet cover story, shown in every episode alongside its other cover story (text reports here; photographs only in image-report conditions); it is not part of the sign test.

Model	Raw + correctness	Reports masked	Outcome masked	Both masked	Correctness only
Development episodes (exploratory)					
Qwen3-VL-8B	.50	.48	.12	.04	.03
InternVL3.5-8B	.48	.27	.12	.02	.01
Held-out episodes					
Qwen3-VL-8B	.44	.40	.08	.01	.02
InternVL3.5-8B	.44	.26	.10	.03	.02
Gemma-3-27B	.87	.65	.18	.04	.00
Qwen3-VL-32B	.60	.60	.13	.02	.00
InternVL3.5-38B	.42	.37	.23	.02	.01

Table 15: Weight on repeating the previous outcome (log-odds) under history masking. Column names specify the information removed; correctness statements remain visible. Development rows are exploratory; held-out masking predictions were fixed before those runs. Absolute held-out accuracies appear in Table 2 and paired accuracy differences with CIs in Fig. 11; they are not repeated here.

Model	Wording	Raw	Raw+c.	Corr. only	Summ.	Summ.−raw [95% CI]	Repeat
Qwen3-VL-8B	Original	.485	.534	.640	.617	+.132 [.108, .153]	.46
	Reworded	.488	.533	.624	.607	+.119 [.093, .144]	.50
	Compact	.563	.561	.580	.597	+.034 [.008, .059]	.32
InternVL3.5-8B	Original	.471	.530	.640	.628	+.157 [.140, .174]	.28
	Reworded	.473	.538	.644	.627	+.153 [.133, .175]	.26
	Compact	.550	.559	.593	.564	+.014 [−.019, .045]	.13

Table 16: Prompt-wording check (development episodes, text reports; accuracy averaged over counterbalanced copies, $t \geq 2$). Repeat: regression weight on the previous outcome under the raw history.

	Raw history, text reports				Correctness-only history			
	Judgment	Choice	Choice follows		Chose first listed		Image: acc. when correct is	
Model	names better	follows better	own judgment	κ	Text	Image	first	second
Qwen3-VL-8B	.633	.523	.579	.09	.455	.523	.571	.525
InternVL3.5-8B	.674	.524	.527	−.02	.526	.803	.847	.243
Gemma-3-27B	.602	.531	.576	.07	.438	.792	.828	.243
Qwen3-VL-32B	.607	.532	.537	.00	.537	.918	.941	.105
InternVL3.5-38B	.672	.633	.691	.12	.524	.903	.920	.114

Table 17: Judgment and report position (held-out; averaged over counterbalanced copies). Names/follows better: the scheduled better advisor; always naming the advisor that is better over most of the episode scores .75. κ: agreement between judgment and choice. With raw histories and image reports, the larger models chose the first-listed image on .88–.93 of trials (text .57–.58).

Masking, prompt-wording checks and the Bayesian reference were added after the commit. Masking predictions followed development results but preceded held-out masking: this later-specified replication is separate from the primary test. API behavioural comparisons, judgment–choice and image-position checks, stale-value regression and readout

Model	Checkpoint	Revision
Qwen3-VL-8B	Qwen/Qwen3-VL-8B-Instruct	0c351dd01ed87e9c1b53cbc748cba10e6187ff3b
InternVL3.5-8B	OpenGVLab/InternVL3_5-8B-HF	741a7d03020411e666c6109218ab71e08151ef86
Gemma-3-27B	google/gemma-3-27b-it	005ad3404e59d6023443cb575daa05336842228a
Qwen3-VL-32B	Qwen/Qwen3-VL-32B-Instruct	0cfaf48183f594c314753d30a4c4974bc75f3ccb
InternVL3.5-38B	OpenGVLab/InternVL3_5-38B-HF	7c830fc25e874f9a861087e3c70587f2b9b555f3

Table 18: Checkpoints and pinned revisions (resolved 2026-11-11). B in model names denotes billions of parameters.

Model	Raw	Correctness-only	Summary	Summary − raw	Correctness-only − raw
Qwen3-VL-8B	49.27	58.65	59.41	10.15 [9.03, 11.21]	9.38 [8.21, 10.50]
InternVL3.5-8B	48.99	61.00	61.88	12.89 [11.91, 13.88]	12.01 [10.71, 13.24]
Gemma-3-27B	47.15	56.25	55.94	8.79 [7.29, 10.25]	9.10 [7.67, 10.53]
Qwen3-VL-32B	48.31	58.40	62.16	13.86 [13.13, 14.58]	10.09 [9.06, 11.14]
InternVL3.5-38B	53.90	61.68	63.14	9.24 [8.39, 10.05]	7.78 [6.86, 8.68]

Table 19: Per-copy scoring on the full held-out set. Accuracy is in percent and paired differences in percentage points, with 95% CIs from 10,000 bootstrap resamples of 48 episode pairs. Each copy is scored before averaging; the original counterbalanced-logit estimates appear in Table 1.

			Agreement with strategy (κ)					Quota vs. learner	Since reversal	
Model	History	Acc.	Repeat	WSLS	Count	Last 5	Learner	follows learner	1–3	8–10
Claude Fable 5.2	Raw	.629	.02	.43	.12	.79	.87	.96	.06	.94
	Summary	.585	.02	.05	1.00	.11	.13	.82	.50	.49
GPT-6-Astra	Raw	.628	−.03	.41	.09	.77	.78	.91	.09	.99
	Summary	.585	.02	.05	1.00	.11	.13	.82	.50	.49
Open models (5)	Raw	.467–.503	.08–.31	.00–.04	.01–.05	−.01–.06	.01–.08	.59–.80	.47–.56	.43–.51
	Summary	.533–.590	.01–.02	.01–.07	.04–.68	.09–.16	.08–.17	.67–.81	.47–.51	.51–.58
Rule alone		repeat .458; WSLS .623; count .585; last 5 .679; learner .629; outcome quota (hidden) .702								

Table 20: Behaviour on the API subset (24 episode pairs, $t = 4, 8, \ldots, 80$; per-copy choices for all models). Strategy definitions follow Fig. 9; Learner denotes the Bayesian volatility learner. Quota vs. learner: share following the learner when it disagrees with the hidden outcome-quota rule. Since reversal: proportion of discrete choices following the currently better advisor in trials 1–3 or 8–10 of a reversed block (offsets 0–2 or 7–9). Only genuine reversals are included; initial blocks are excluded. These windows match Fig. 4, which plots choice probabilities instead.

	Candidate mass			Format
Model	Mean	Min	Share < 0.5	following
Qwen3-VL-8B	1.000	.422	.000	1.000
InternVL3.5-8B	.997	.177	.000	1.000
Gemma-3-27B	.868	.000	.125	.875
Qwen3-VL-32B	.996	.807	.000	1.000
InternVL3.5-38B	1.000	.684	.000	1.000

Table 21: Readout quality pooled over main-task histories and report types on held-out episodes. Candidate mass sums the two answer-token probabilities; format following is the proportion of greedy outputs equal to an answer word. Values are rounded. Gemma's failures concentrate in summarized and chart histories, as detailed in the text.

sensitivity were specified after inspecting held-out outputs and are exploratory.

An exploratory development-only test of an open reasoning checkpoint was completed but excluded because truncation, parsing sensitivity and sparse sampling limit interpretation.

Pilot disclosure. The earlier RTX 3090 pilot used fixed advisor-introduction order, removed in the current episodes. Outcome repetition, image-position effects and poor chart reading motivated the controls and readability criterion. These pre-plan outputs are excluded from main-task estimates and supply no held-out evidence.